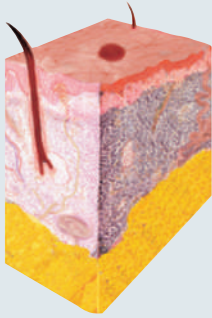
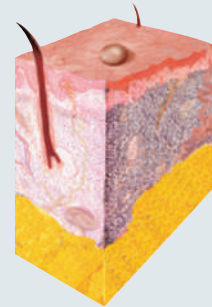
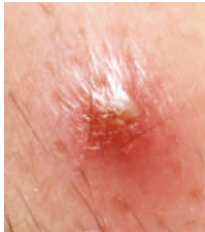
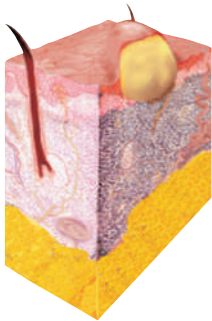


Macule (MAK-yuhl) (plural: maculae [MAK-yuh-ly])	 © wk1003mike/Shutterstock.com.		Flat spot or discoloration on the skin.	Freckle or “liver” spot.
Nodule (NAHD-yul)	 © Sue McDonald/Shutterstock.com.		A solid bump larger than 0.4 inches (1 cm) that can be easily felt. Requires medical referral.	Swollen lymph nodes, rheumatoid nodules.
Papule (PAP-yool)	 © Ocskay Bence/Shutterstock.com.		A small elevation on the skin that contains no fluid, but may develop pus.	Acne, warts, elevated nevi.
Pustule (PUS-chool)	 © Faiz Zaki/Shutterstock.com.		Raised, inflamed, papule with a white or yellow center containing pus in the top of the lesion.	Acne, impetigo, folliculitis.
Tumor (TOO-mur)	 © Courtesy DermNet NZ		Abnormal mass varying in size, shape, and color. Any type of abnormal mass, not always cancer. Requires medical referral.	Cancer.

(Continues)

table 9-1
(Continued)

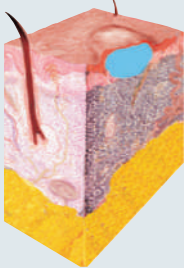
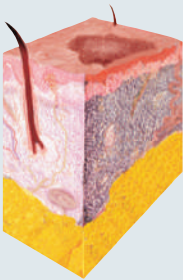
Primary Lesion	Image	Graphic	Description	Examples
Vesicle (VES-ih-kuhl)			Small blister or sac containing clear fluid, lying within or just beneath the epidermis. Requires medical referral if cause is unknown or untreatable with over-the-counter products.	Poison ivy, poison oak.
Wheal (WHEEL)	 © Margoe Edwards/Shutterstock.com		An itchy, swollen lesion that can be caused by a blow, scratch, bite of an insect, or urticaria (skin allergy), or the sting of a nettle. Typically resolves on its own, but referral to a physician should be considered when the condition lasts more than 3 days.	Hives, mosquito bites.

table 9-2
SECONDARY LESIONS

Secondary Lesion	Image	Graphic	Description	Examples
Crust (KRUST)	 © Pan Xunbin/Shutterstock.com		Dead cells that form over a wound or blemish while healing; accumulation of sebum and pus, sometimes mixed with epidermal cells.	Scab, sore.
Excoriation (ek-skor-ee-AY-shun)	 R. Baran "The Nail in Differential Diagnosis" with permission of Informa (London).		Skin sore or abrasion produced by scratching or scraping.	Nail cuticle damage from nail biting.

Fissure (FISH-ur)	 © Ilbrakv/Shutterstock.com.		Crack in the skin that penetrates the dermis.	Severely cracked or chapped hands/lips.
Keloid (KEE-loyd)			A thick scar resulting from excessive growth of fibrous tissue that will form along any type of scar for people susceptible to these.	
Scale (SKAYL)	 © Christine Langer-Pueschel/Shutterstock.com.		Thin, dry, or oily plate of epidermal flakes.	Excessive dandruff, psoriasis.
Scar (skahr) or cicatrix (SIK-uh-triks)	 © Geo-grafika/Shutterstock.com.		Slightly raised or depressed mark on the skin formed after an injury or a lesion of the skin has healed.	Postoperative repair.
Ulcer (UL-sur)	 © Ilya Andriyanov/Shutterstock.com.		Open lesion on the skin or mucous membrane of the body, accompanied by loss of skin depth and possibly weeping of fluids or pus. Requires medical referral, particularly for clients with underlying medical conditions, such as diabetes.	Chicken pox, herpes.

Discuss Disorders of the Sebaceous and Sudoriferous Glands

After reading this section, you will be able to:

- LO 4 Describe common skin inflammations and infections.
- LO 5 List and describe disorders of the sebaceous and sudoriferous glands.

figure 9-6
Comedone.



figure 9-7
Milia.



figure 9-8
Acne.



There are several common disorders of the sebaceous (oil) glands that the barber should be able to understand and identify.

An **open comedo**, also known as a *blackhead*, is a hair follicle filled with keratin and sebum. **Comedones** (KAHM-uh-dohnz) appear most frequently on the face, especially in the T-zone, the center of the face (Figure 9-6). When the sebum of the comedo is exposed to the environment, it oxidizes and turns black. When the follicle is closed and not exposed to the environment, the sebum remains a white or cream color and is a **closed comedo**, also known as a **whitehead** (WHYT-hed), and appears as a small bump just under the skin surface.

Milia (MIL-ee-uh) are benign, keratin-filled cysts that appear just under the epidermis and have no visible opening. They resemble small sesame seeds and are almost always perfectly round. They are commonly associated with newborn babies but can appear on the skin of people of all ages. They are usually found around the eyes, cheeks, and forehead, and appear as small, whitish masses (Figure 9-7).

Acne (AK-nee), also known as *acne vulgaris*, is a skin disorder characterized by chronic inflammation of the sebaceous glands from retained secretions (Figure 9-8). Propionibacterium acne occurs when obstructed follicles are deprived of oxygen, which allows acne bacteria to multiply. Fatty acids obtained from sebum in the follicle are the main food source for acne bacteria. The bacteria multiply, causing inflammation and swelling in the follicle and eventually rupture the follicle wall. When the wall ruptures, the immune system causes blood to rush to the ruptured follicle, carrying white blood cells to fight the bacteria. An acne papule is an inflammatory acne lesion resulting from this rupture of the follicle wall and the infusion of blood. A pustule forms from the papule when enough white blood cells accumulate to form pus, which is primarily composed of dead white blood cells. Acne conditions are rated in four grades:

- **Grade I.** Minor breakouts, mostly open comedones, some closed comedones, and a few papules and pustules
- **Grade II.** Many closed comedones, more open comedones, and more papules and pustules
- **Grade III.** Redness and inflammation with many papules and pustules
- **Grade IV.** Cysts with comedones, papules, pustules, and inflammation (cystic acne)

A **sebaceous cyst** is a large, protruding pocket-like lesion filled with sebum. Sebaceous cysts are frequently seen on the scalp and the back and may be surgically removed by a dermatologist.

Courtesy Mark Lees Skin Care, Inc.

www.dermnet.com.

© Suzanne Tucker/Shutterstock.com.

Seborrheic dermatitis (seb-oh-REE-ick dur-muh-TY-tis) or seborrhea, is a skin condition caused by an inflammation of the sebaceous glands, and is often characterized by redness, dry or oily scaling, crusting, and/or itchiness (**Figure 9-9**). Excessive oiliness on the skin or scalp may also indicate the presence of seborrhea. The red, flaky skin often appears in the eyebrows and beard, in the scalp and hairline, at the middle of the forehead, and along the sides of the nose. Dandruff and cradle cap are forms of seborrheic dermatitis that are sometimes treated with topical creams.

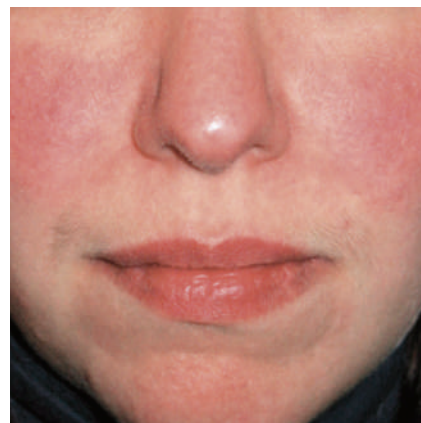
Rosacea (roh-ZAY-see-uh), previously called *acne rosacea*, is a chronic condition that appears primarily on the cheeks and nose (**Figure 9-10**). It is characterized by flushing (redness), **telangiectasis** (tel-an-jee-EK-tuh-sus) (distended or dilated surface blood vessels), and, in some cases, the formation of papules and pustules. The cause of rosacea is unknown, but the condition is thought to be genetic. Certain factors are known to aggravate the condition in some individuals. These factors include exposure to heat, sun, and very cold weather; ingestion of spicy foods, caffeine, and alcohol; and stress. Rosacea can be treated and kept under control by using medication prescribed by a physician, using proper skin care products designed for especially sensitive skin, and avoiding the aggravating flare factors listed above.

Asteatosis (as-tee-ah-TOH-sis) is a condition of dry, scaly skin, characterized by the absolute or partial deficiency of sebum. It can be the result of old age, exposure to cold or alkalis, or bodily disorders. **Steatoma** (stee-ah-TOH-muh): a sebaceous cyst or fatty tumor that is filled with sebum. It is a subcutaneous tumor of the sebaceous glands that can range in size from a pea to an orange. A steatoma usually occurs on the scalp, neck, or back and is sometimes called a *wen*.

figure 9-9
Seborrheic dermatitis.



figure 9-10
Rosacea.



ACNE TREATMENT

Minor forms of acne can be treated without medical referral. Following are the basics of acne treatment:

- Daily use of gentle cleansers, formulated for a specific skin type. The use of harsh cleansers can make skin too dry and sebaceous glands will generate more sebum, creating an even bigger problem. These foamy, rinse-off products remove dirt, debris, and excess oil from the skin. Toners may be helpful for customers with excessively oily skin.
- Follicle exfoliants are leave-on products that help to remove cell buildup from the follicles, allowing oxygen to penetrate the follicles, killing bacteria. Commonly used ingredients in these products are alpha hydroxy acid, salicylic acid, and benzoyl peroxide. Benzoyl peroxide can be especially effective since it helps to shed cellular debris and also kills the acne bacteria. These products are generally not used all over because of their drying properties and are only used as a spot treatment. Clients should be warned about possible allergic reactions to these products and should stop using them if clients experience numbness in the area or swelling of the hands, feet, or mouth.
- Use of a light moisturizer to keep skin balanced and reduce the risk of excess sebum production can be helpful.

IDENTIFY DISORDERS OF THE SUDORIFEROUS GLANDS

There are several disorders of the sudoriferous (sweat) glands that barbers should be able to identify and describe.

- **Anhidrosis** (an-hy-DROH-sis) is a deficiency in perspiration or the inability to sweat, often a result of damage to autonomic nerves. This condition can be life threatening and requires medical attention.
- **Bromhidrosis** (broh-mih-DROH-sis) results in foul-smelling perspiration, usually noticeable in the armpits or on the feet. This condition is generally caused by bacteria; severe cases require medical referral.
- **Hyperhidrosis** (hy-pur-hy-DROH-sis) is characterized by excessive sweating, caused by heat or general body weakness. This condition requires medical referral.
- **Miliaria rubra** (mil-ee-AIR-ee-ah ROOB-rah), also known as *prickly heat*, is an acute inflammatory disorder of the sweat glands, characterized by the eruption of small red vesicles accompanied by burning, itching skin. It is caused by exposure to excessive heat and usually clears in a short time without treatment.

RECOGNIZE COMMON INFLAMMATIONS AND INFECTIONS OF THE SKIN

As a barber, you will need to be able to distinguish between noncontagious and contagious skin inflammations and infections.

- **Dermatitis** (dur-muh-TY-tis) is the general term for an inflammatory condition of the skin. The lesions may appear in various forms, such as vesicles or papules. **Irritant contact dermatitis**, abbreviated ICD, occurs when irritating substances, such as chemicals or tints, temporarily damage the epidermis. Unlike allergic contact dermatitis, irritant contact dermatitis is not usually chronic if precautions are taken, such as wearing gloves when working with chemicals. This condition is also called **dermatitis venenata** (dur-muh-TY-tis VEN-uh-nah-tuh).
- **Eczema** (EG-zuh-muh) is an inflammatory skin disease that may be acute or chronic in nature and present in many forms of dry or moist lesions. Eczema is frequently accompanied by itching or burning and all cases should be referred to a physician for treatment. Eczema is not contagious and its cause is unknown (**Figure 9-11**).
- **Psoriasis** (suh-RY-uh-sis) is a chronic inflammatory skin disease characterized by dry red patches covered with coarse, silvery scales. Psoriasis usually occurs on the scalp, elbows, knees, chest, or lower back, but rarely on the face. If irritated, bleeding points can occur. It is not contagious and its cause is unknown.
- **Herpes simplex I** (HER-pee-z SIM-pleks ONE) is a recurring viral infection that produces fever blisters or cold sores characterized by a single vesicle

figure 9-11
Eczema.



or group of vesicles with red, swollen bases. The blisters usually appear on the lips, nostrils, or other parts of the face, and rarely last more than a week. Herpes simplex II is caused by the same virus and designates occurrences below the waist. Both forms of herpes simplex are contagious and require medical treatment (**Figure 9-12**).

- **Ivy dermatitis** (dur-muh-TY-tis) is a skin inflammation caused by exposure to poison ivy, poison oak, or poison sumac leaves. Blisters and itching develop soon after contact occurs. The condition may spread to other parts of the body by contact with contaminated hands, clothing, objects, or anything that was exposed to the plant itself. If the plant oil, which can be irritating, remains on the skin, it also can be spread from one person to another by direct contact. Serious cases should be referred to a physician.

figure 9-12
Herpes simplex I.



Recognize Pigment Disorders and Hypertrophies of the Skin

After reading this section, you will be able to:

- LO 6** List and describe types of skin pigmentations.
- LO 7** Identify common skin hypertrophies.

Barbers are often in the position to notice changes in a client's skin and it is important to be able to differentiate between skin pigmentations and hypertrophies. For example, a flat mole may look like a dark freckle, but if you see a change in the color or growth of the mole, bring it to your client's attention for medical follow up.

IDENTIFY TYPES OF SKIN PIGMENTATIONS

Skin pigment can be affected by internal factors such as heredity or hormonal fluctuations, or by external factors such as prolonged exposure to the sun. Abnormal colorations, known as **dyschromias** (dis-chrome-ee-uhs), accompany skin disorders and are symptoms of many systemic disorders. A change in pigmentation can also be observed when certain medications are being taken, because of photosensitive reactions related to use of certain antibiotics. The following conditions relate to changes in the pigmentation of the skin:

- **Hyperpigmentation** (hy-pur-pig-men-TAY-shun) means darker than normal pigmentation, appearing as dark splotches.
- **Hypopigmentation** (hy-poh-pig-men-TAY-shun) is the absence of pigment, resulting in light or white splotches.

DID YOU KNOW?

Skin cancer is preventable and early detection is possible, if you know what to look for. Be aware of the following as you service your clients:

- Any unusual lesions on the skin or on the scalp or change in an existing lesion or mole
- Melanomas, irregularly shaped dark spots, sometimes found on the scalp and ears and often first detected by a barber, who can see the back of the head and behind the ears!
- A new lesion or discoloration on the skin or scalp
- A client who complains about sores that do not heal or unexpected skin bleeding
- Recurrent scaly areas that may be rough to the touch, especially in sun-exposed areas such as the face, arms, or hands

Always discuss prevention with every client. If you become aware of any of these conditions, suggest that your client consult a physician.

figure 9-13
Albinism.



figure 9-14
Port-wine stain.



figure 9-15
Vitiligo.



- **Albinism** (AL-bi-niz-em) is congenital hypopigmentation, or absence of melanin pigment in the body, including the skin, hair, and eyes (Figure 9-13). Hair is silky white. The skin is pinkish white and will not tan. The eyes are pink, and the skin is sensitive to light and ages early.
- **Chloasma** (kloh-AZ-ma), also known as the *mask of pregnancy*, is a condition characterized by hyperpigmentation on the skin in spots that are not elevated. They are generally caused by cumulative sun exposure and can be helped by exfoliation or treated by a dermatologist.
- **Lentigines** (len-TIJ-e-neeZ) (singular: lentigo [len-TY-goh]) is the technical term for freckles, small yellow-colored to brown-colored spots on skin exposed to sunlight and air. It is also commonly referred to as *liver spots* in older adults, although there is no relationship to the liver.
- **Leukoderma** (loo-koh-DUR-muh) is a skin disorder characterized by light abnormal patches (hypopigmentation); it is caused by a burn, scar, inflammation, or congenital disease that destroys the pigment-producing cells. Examples are vitiligo and albinism.
- **Nevus** (NEE-vus), also known as a *birthmark*, is a small or large malformation of the skin due to abnormal pigmentation or dilated capillaries.
- **Stain** (STAYN) is an abnormal brown-colored or wine-colored skin discoloration with a circular or irregular shape (Figure 9-14). Its permanent color is due to the presence of darker pigment. Stains can be present at birth, or they can appear during aging, after certain diseases, or after the disappearance of moles, freckles, and liver spots. The cause is often unknown.
- **Tan** (TAN) is the change in pigmentation of skin caused by exposure to the sun or ultraviolet light.
- **Vitiligo** (vi-til-EYE-goh) is a hereditary condition that causes hypopigmented spots and splotches on the skin that often appear milky white. Recent research suggests that this disorder is part of an autoimmune disease (Figure 9-15). Skin with vitiligo must be protected from overexposure to the sun.

DESCRIBE HYPERTROPHIES OF THE SKIN

As a barber, you can expect to see some common hypertrophies on your clients' face, scalp, or neck during a skin or scalp analysis. An **hypertrophy** (hy-pur-troh-fee) of the skin is an abnormal growth of the skin. Many hypertrophies are benign (harmless) however, if you see any changes occur, be sure to share your observation with your client. Examples of common hypertrophies seen in the barbershop are as follows:

- A **keratoma** (kair-uh-TOH-muh) is an acquired, superficial, thickened patch of epidermis. A callus is a keratoma that is caused by continued, repeated pressure or friction on any part of the skin, especially the hands and feet. If the thickening grows inward, it is called a *corn*.
- A **mole** (MOHL) is a small brownish spot or blemish on the skin, ranging in color from pale tan to brown or bluish black. Some moles are small and flat, resembling freckles; others are raised and darker in color. Large dark hairs often occur in moles. Any change in a mole requires medical attention.

<http://www.dermnet.com>

Courtesy Mark Lees Skin Care, Inc.

- A **skin tag** is a small brown-colored or flesh-colored outgrowth of the skin. Skin tags occur most frequently on the neck and chest and can be easily removed by a dermatologist.
- A **verruca** (vuh-ROO-kuh), also known as a **wart**, is an hypertrophy of the papillae and epidermis. It is caused by a virus and is infectious. Verruca can spread from one location to another, particularly along a scratch in the skin. A dermatologist can be helpful in removing and reducing the recurrence of warts.


CAUTION

Do not treat or remove hair from moles.

Understand Skin Cancer

After reading this section, you will be able to:

LO8 Identify and describe types of skin cancer.

Skin cancer has become one of the most common cancers. It is also becoming one of the most common causes of cancer-related deaths because of general complacency about prevention and a lack of knowledge about the signs and real risks, particularly in young people. Barbers should recognize the signs of potential skin cancer and always refer clients. In this case, “better safe than sorry” is absolutely true! Do not let someone’s young age or general good health stop you from being the one who saves someone’s life through early diagnosis and treatment. There are three types of skin cancer (see Table 9-3).

table 9-3
TYPES OF SKIN CANCER

Cancer Type	Description	Image
Basal cell carcinoma (BAY-zul SELL kahr-sin-OH-muh)	Most common and least severe skin cancer; characterized by light or pearly nodules and has a 90% survival rate with early diagnosis and treatment.	
Squamous cell carcinoma (SKWAY-mus SELL kahr-sin-OH-muh)	More serious than basal cell carcinoma; characterized by scaly red papules or nodules. It can spread to other parts of the body; survival rates depend on the stage at diagnosis.	
Malignant melanoma (muh-LIG-nent mel-ahn-OH-muh)	Least common of the cancers, but is 100% fatal if left untreated—early detection and treatment can result in a 94% 5-year survival rate, which drops drastically, to 62%, once it reaches local lymph nodes; characterized by black or dark brown patches on the skin that may appear uneven in texture, jagged, or raised.	



DID YOU KNOW?

The American Cancer Society recommends using the ABCDE Cancer Checklist to help make potential skin cancer easier to recognize. When checking for existing lesions or hypertrophies, look for changes in any of the following: A, asymmetry; B, border; C, color; D, diameter; or E, evolution. A physician should examine any of these changes.

Clients should be advised to regularly see a dermatologist for checkups of the skin, especially if any changes in coloration, size, or shape of a mole are detected, if the skin bleeds unexpectedly, or if a lesion or scrape does not heal quickly.

Home self-examinations can also be an effective way to check for signs of potential skin cancer between scheduled doctor visits. You should advise clients to check for—as part of their self-care exam—any changes in existing moles and pay attention to any new visible growths on the skin. Clients should also be advised to ask a spouse, friend, or loved one to check areas they cannot adequately see on a routine basis. These areas would include the back, scalp, and around the ears.

If detected early, anyone with these three forms of skin cancer may be cured. Barbers serve a unique role by being able to recognize the appearance of serious skin disorders and referring the client to a dermatologist for diagnosis and treatment.

Know How to Maintain the Health of Your Skin

Diet and protection are the major factors involved in maintaining the skin's overall health and appearance. Proper and beneficial dietary choices, as well as protecting the skin, help regulate hydration and oil production and optimize the function of cells.

- **Foods.** Eating a well-balanced diet of the three basic food groups of fats, carbohydrates, and proteins is the best way to support the health of the skin.
- **Vitamins and supplements.** Various nutrients aid in healing, softening, and fighting diseases of the skin. Vitamin A supports the overall health of the skin, vitamin C is important to skin and tissue repair, vitamin D promotes healthy and rapid healing of the skin, and vitamin E helps to fight against the harmful effects of the sun's rays. You should always suggest that customers speak to their physician prior to starting any regimen of supplements.
- **Water.** Ingesting plenty of fluid sustains the health of the cells, aids in the elimination of toxins and waste, helps to regulate the body's temperature, and aids in proper digestion.
- **Protection.** Using sunscreen with an SPF30 when being in the sun is necessary, as is avoiding sun exposure when possible, both of which will reduce the risk of cell and tissue damage, which leads to premature aging and possibly cancer. Use moisturizers to assist the surface of the skin to stay soft and supple when in dry environments, such as arid locations, or long-term exposure to water and drying agents.



DID YOU KNOW?

Some factors that influence aging of skin cannot be controlled (intrinsic factors), for example, heredity, gravitational pull, and repeated use of muscles. Many factors can be controlled (extrinsic factors), for example, exposure to sun, alcohol and tobacco use, poor nutrition, and stress.

REVIEW QUESTIONS

- 1 Briefly describe healthy skin.
- 2 Name the two main divisions of the skin and describe the layers within each division.
- 3 Identify the appendages of the skin.
- 4 How is the skin nourished?
- 5 Name three types of nerve fibers found in the skin.
- 6 What determines the color of the skin?
- 7 Identify two types of glands found in the skin and describe their functions.
- 8 List the six important functions of the skin.
- 9 What is a lesion?
- 10 What are the characteristics of primary skin lesions?
- 11 Describe the characteristics of secondary skin lesions.
- 12 List the characteristics of the following: eczema, herpes simplex, psoriasis, irritant contact dermatitis (dermatitis venenata).
- 13 What are some characteristics of seborrheic dermatitis?
- 14 Which three disorders of the sudoriferous glands requires medical attention?
- 15 Define a primary skin lesion and list ten types.
- 16 Define a secondary skin lesion and list seven types.
- 17 Name and describe at least five disorders of the sebaceous glands.
- 18 Name and describe at least five changes in skin pigmentation.
- 19 Name and describe the three forms of skin cancer.
- 20 What is the most common type of skin cancer?

CHAPTER GLOSSARY

acne (AK-nee)	p. 234	skin disorder characterized by chronic inflammation of the sebaceous glands from retained secretions
adipose tissue (AD-uh-pohs TISH-oo)	p. 225	a technical term for fat; gives smoothness and contour to the body
albinism (AL-bi-niz-em)	p. 238	congenital leukoderma, or absence of melanin pigment in the body

anhidrosis (an-hy-DROH-sis)	p. 236	a deficiency in perspiration or the inability to sweat, often a result of damage to autonomic nerves
asteatosis (as-tee-ah-TOH-sis)	p. 235	a condition of dry, scaly skin, characterized by the absolute or partial deficiency of sebum
basal cell carcinoma (BAY-zul SELL kahr-sin-OH-muh)	p. 239	most common and least severe type of skin cancer
blackhead (BLAK-hed)	p. 228	an open comedone; consists of an accumulation of excess oil (sebum) that has been oxidized to a dark color
bromhidrosis (bro-mih-DROH-sis)	p. 236	foul-smelling perspiration
bullae (BULL-uh)	p. 230	large blister containing a watery fluid
chloasma (kloh-AZ-ma)	p. 238	non-elevated spots due to increased pigmentation in the skin
closed comedo	p. 234	also known as a <i>whitehead</i> ; a small bump just under the skin surface with white or cream colored sebum
collagen (KAHL-uh-jen)	p. 226	fibrous protein that gives the skin form and strength
comedone (KAHM-uh-dohn)	p. 234	a mass of hardened sebum and skin cells in a hair follicle that may be open (blackhead) or closed (whitehead)
crust (KRUST)	p. 232	dead cells that have accumulated over a wound while healing
cyst (SIST)	p. 230	a closed, abnormally developed sac containing fluid or morbid matter, above or below the skin
derma (DURM-uh)	p. 224	technical name for skin; also another name for the dermis
dermatitis (dur-muh-TY-tis)	p. 236	an inflammatory condition of the skin
dermatitis venenata (dur-muh-TY-tis VEN-uh-nah-tuh)	p. 236	also known as <i>irritant contact dermatitis</i> ; an eruptive skin condition due to contact with irritating substances such as tints or chemicals
dermatology (dur-muh-TAHL-uh-jee)	p. 222	a branch of medical science that deals with the study of the skin
dermis (DUR-mis)	p. 224	second or inner layer of the skin; also known as the <i>derma</i> , <i>corium</i> , <i>cutis</i> , or <i>true skin</i>
dyschromias (dis-chrome-ee-uhs)	p. 237	abnormal skin colorations
eczema (EG-zuh-muh)	p. 236	inflammatory skin condition characterized by painful itching; dry or moist lesion forms
elastin (ee-LAS-tin)	p. 226	protein base similar to collagen that forms elastic tissue
epidermis (ep-ih-DUR-mus)	p. 223	outermost layer of the skin; also called the <i>cuticle</i> or <i>scarf skin</i>
excoriation (ek-skor-ee-AY-shun)	p. 232	skin sore or abrasion caused by scratching or scraping
fissure (FISH-ur)	p. 233	a crack in the skin that penetrates to the dermis

herpes simplex I (HER-pee-z SIM-pleks ONE)	p. 236	fever blister or cold sore; a recurring viral infection
hyperhidrosis (hy-pur-hy-DROH-sis)	p. 236	excessive perspiration or sweating
hyperpigmentation (hy-pur-pig-men-TAY-shun)	p. 237	darker than normal pigmentation, appearing as dark splotches
hypertrophy (hy-pur-troh-fee)	p. 238	abnormal skin growth
hypopigmentation (hy-poh-pig-men-TAY-shun)	p. 237	the absence of pigment, resulting in light or white splotches
irritant contact dermatitis	p. 236	abbreviated ICD; also known as <i>dermatitis venenata</i> ; occurs when irritating substances temporarily damage the epidermis
ivy dermatitis (EYE-vee dur-muh-TY-tis)	p. 237	a skin inflammation caused by exposure to poison ivy, poison oak, or poison sumac
keloid (KEE-loyd)	p. 233	thick scar resulting from excessive tissue growth
keratoma (kair-uh-TOH-muh)	p. 238	technical name for a callus, caused by pressure or friction
lentigines (len-TIJ-e-nee-z)	p. 238	technical name for freckles
lesion (LEE-zhun)	p. 230	a structural change in the tissues caused by injury or disease
leukoderma (loo-koh-DUR-muh)	p. 238	skin disorder characterized by abnormal white patches
macule (MAK-yuhl)	p. 231	spot or discoloration of the skin, such as a freckle
malignant melanoma (muh-LIG-nent mel-ahn-OH-muh)	p. 239	most severe form of skin cancer
melanin (MEL-uh-nin)	p. 226	coloring matter or pigment of the skin; found in the stratum germinativum of the epidermis and in the papillary layers of the dermis
milium (MIL-ee-ah)	p. 234	technical name for milk spots; small, benign, whitish bumps that occur when dead skin is trapped in the surface of the skin; commonly seen in infants
miliaria rubra (mil-ee-AIR-ee-ah ROOB-rah)	p. 236	also known as <i>prickly heat</i> , is an acute inflammatory disorder of the sweat glands, characterized by the eruption of small red vesicles accompanied by burning, itching skin
mole (MOHL)	p. 238	small brownish spot on the skin
motor nerve fibers (MOH-tur NURV FY-buhrs)	p. 225	nerve fibers distributed to the arrector pili muscles, which are attached to the hair follicles
nevus (NEE-vus)	p. 238	technical name for a birthmark
nodule (NAHD-yul)	p. 231	a solid bump larger than 0.4 inches (1 cm) that can be easily felt
open comedo	p. 234	also known as a <i>blackhead</i> ; a hair follicle filled with keratin and sebum
papillary layer (PAP-uh-lair-ee LAY-ur)	p. 225	outer layer of the dermis, directly beneath the epidermis

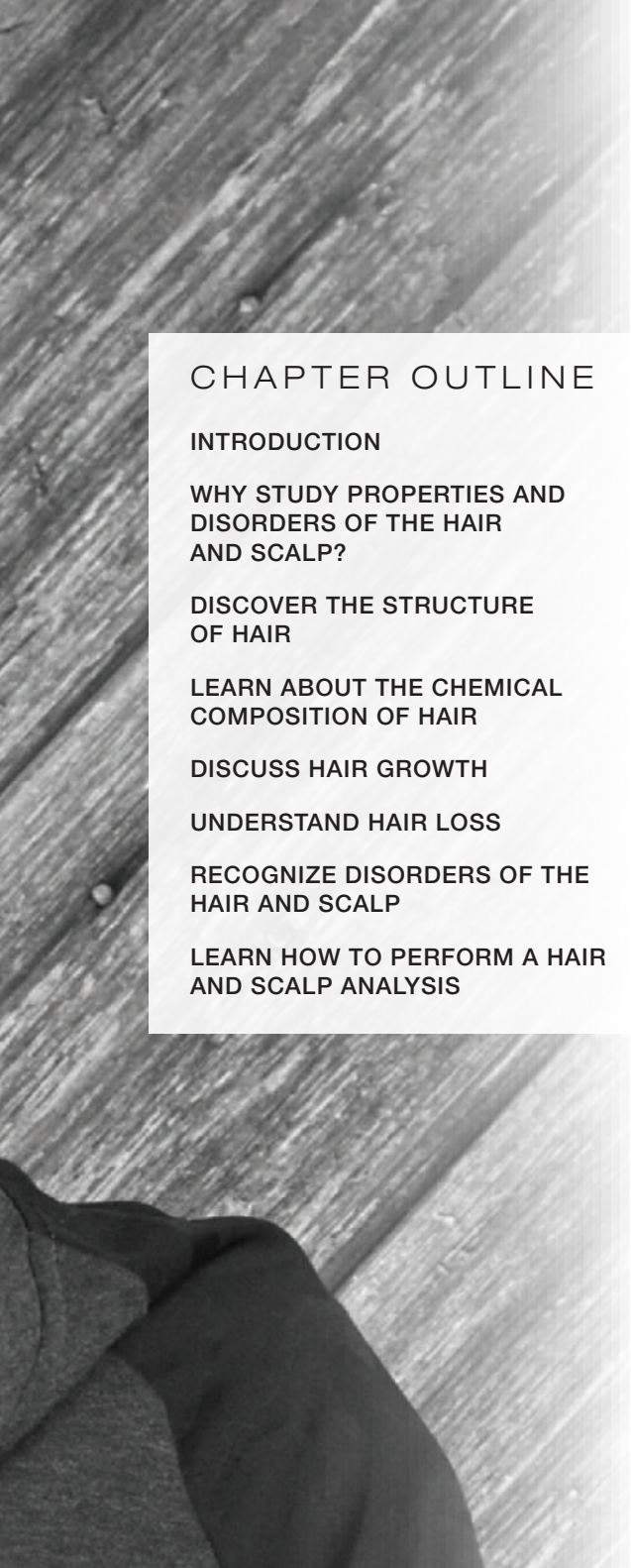
papule (PAP-yool)	p. 231	a small elevation on the skin that contains no fluid, but may develop pus
primary skin lesions	p. 230	lesions that are a different color than the color of the skin or lesions that are raised above the surface of the skin
psoriasis (suh-RY-uh-sis)	p. 236	skin disease characterized by red patches and silvery-white scales
pustule (PUS-chool)	p. 231	raised, inflamed, papule with a white or yellow center containing pus in the top of the lesion
reticular layer (ruh-TIK-yuh-lur LAY-ur)	p. 225	deeper layer of the dermis
rosacea (roh-ZAY-see-uh)	p. 235	chronic congestion of the skin characterized by redness, blood vessel dilation, papules, and pustules
scale (SKAYL)	p. 233	an accumulation of dry or greasy flakes on the skin
scar (skahr) or cicatrix (SIK-uh-triks)	p. 233	slightly raised or depressed mark on the skin formed after an injury or a lesion of the skin has healed
sebaceous cyst	p. 234	a large, protruding pocket-like lesion filled with sebum, frequently seen on the scalp and back
sebaceous glands (sih-BAY-shus GLANDZ)	p. 228	oil glands of the skin connected to hair follicles
seborrheic dermatitis (seb-oh-REE-ick dur-muh-TY-tis)	p. 235	is a skin condition caused by an inflammation of the sebaceous glands, and is often characterized by redness, dry or oily scaling, crusting, and/or itchiness
sebum (SEEB-um)	p. 228	an oily substance secreted by the sebaceous glands
secondary skin lesions	p. 230	lesions characterized by an accumulation of material on the skin surface, such as a crust or scab, or by depressions in the skin surface
secretory nerve fibers (seh-KRUH-toh-ree NURV FY-buhrs)	p. 225	nerves regulate the excretion of perspiration from the sweat glands and the flow of sebum from the oil glands
sensory nerve fibers (SEN-soh-ree NURV FY-buhrs)	p. 225	nerves that react to heat, cold, touch, pressure, and pain and send messages to the brain
skin tag	p. 239	a small brown-colored or flesh-colored outgrowth of the skin occurring most frequently on the neck and chest
squamous cell carcinoma (SKWAY-mus SELL kahr-sin-OH-muh)	p. 239	type of skin cancer more serious than basal cell carcinoma, but not as serious as malignant melanoma
stain (STAYN)	p. 238	abnormal brown- or wine-colored skin discoloration
steatoma (stee-ah-TOH-muh)	p. 235	sebaceous cyst or fatty tumor
stratum corneum (STRAT-um KOR-nee-um)	p. 223	outermost layer of the epidermis; the horny layer
stratum germinativum (STRAT-um jur-min-ah-TIV-um)	p. 224	innermost layer of the epidermis, also known as the <i>basal</i> or <i>Malpighian layer</i>
stratum granulosum (STRAT-um gran-yoo-LOH-sum)	p. 224	granular layer of the epidermis beneath the stratum lucidum; the grainy layer

stratum lucidum (STRAT-um LOO-sih-dum)	p. 223	clear layer of the epidermis, directly beneath the stratum corneum
stratum spinosum (STRAT-um spy-NOH-sum)	p. 224	spiny layer of the epidermis, often considered part of the stratum germinativum
subcutaneous tissue (sub-kyoo-TAY-nee-us TISH-oo)	p. 225	fatty tissue layer that lies beneath the dermis; also called <i>adipose tissue</i>
sudoriferous glands (sood-uh-RIF-uh-rus GLANDZ)	p. 227	sweat glands of the skin
tan (TAN)	p. 238	darkening of the skin due to exposure to ultraviolet rays
telangiectasis (tel-an-jee-EK-tuh-sus)	p. 235	distended or dilated surface blood vessels
tubercle (TOO-bur-kul)	p. 230	abnormal solid lump above, within, or below the skin
tumor (TOO-mur)	p. 231	abnormal cell mass resulting from excessive multiplication of cells
ulcer (UL-sur)	p. 233	open skin lesion accompanied by pus and loss of skin depth; a deep erosion; a depression in the skin, normally due to infection or cancer
verruca (vuh-ROO-kuh)	p. 239	technical name for a wart
vesicle (VES-ih-kuhl)	p. 232	small blister or sac containing clear fluid
vitiligo (vih-til-EYE-goh)	p. 238	an acquired leukoderma characterized by milky-white spots
wart	p. 239	also known as <i>verruca</i> ; an infectious hypertrophy of the papillae and epidermis caused by a virus
wheal (WHEEL)	p. 232	itchy, swollen lesion caused by insect bites or plant irritations, such as nettle
whitehead (WHYT-hed)	p. 234	a closed comedone; consists of accumulated sebum that remains a whitish color because it does not have a follicular opening for exposure to oxygen

10



PROPERTIES AND
DISORDERS OF THE
HAIR AND SCALP



CHAPTER OUTLINE

INTRODUCTION

WHY STUDY PROPERTIES AND
DISORDERS OF THE HAIR
AND SCALP?

DISCOVER THE STRUCTURE
OF HAIR

LEARN ABOUT THE CHEMICAL
COMPOSITION OF HAIR

DISCUSS HAIR GROWTH

UNDERSTAND HAIR LOSS

RECOGNIZE DISORDERS OF THE
HAIR AND SCALP

LEARN HOW TO PERFORM A HAIR
AND SCALP ANALYSIS

LEARNING OBJECTIVES

After reading this chapter, you will be able to:

LO1

Identify and distinguish the different structures of the hair root.

LO2

Identify and distinguish the three layers of the hair shaft.

LO3

Identify and explain the three types of side bonds of the cortex.

LO4

Name and describe the three phases of the hair growth cycle.

LO5

Identify and define seven types of hair loss.

LO6

Identify and describe two FDA-approved treatments for hair loss.

LO7

Identify and define common hair disorders.

LO8

Define common scalp disorders and identify those requiring medical attention.

LO9

Identify the factors to be observed and considered during a hair and scalp analysis.

Introduction

The scientific study of hair, its disorders, and its care is called **trichology** (trih-KAHL-uh-jee), which comes from the Greek words *trichos* (“hair”) and *ology* (“study of”). As you will learn, the hair we see is a part of the integumentary system that protects the head from heat, cold, and injury; it also serves as a source of adornment and an expression of personal style.

why study

PROPERTIES AND DISORDERS OF THE HAIR AND SCALP?

Barbers should study and have a thorough understanding of the properties and disorders of the hair and scalp because:

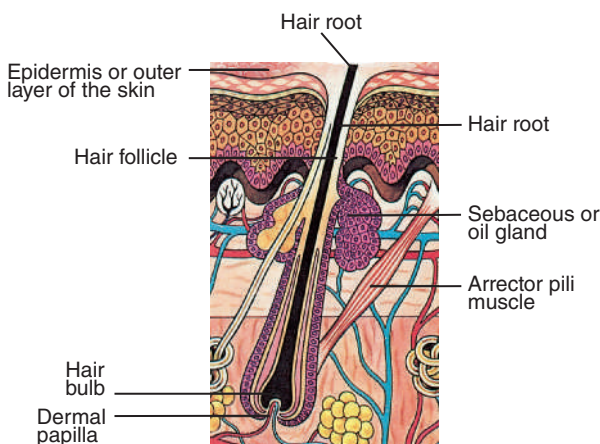
- Barbers need to know how products and chemicals affect the structure of the hair.
- Barbers need to know how to keep a client’s hair and scalp in a healthy condition and how to differentiate between normal and abnormal conditions.
- Barbers need to be able to perform a scalp and hair analysis before performing a service.

Discover the Structure of Hair

After reading this section, you will be able to:

- LO 1 Identify and distinguish the different structures of the hair root.
- LO 2 Identify and distinguish the three layers of the hair shaft.

figure 10-1
Structures of the hair.



Hair is an appendage of the skin, in the form of a slender, threadlike outgrowth of the skin and scalp. It is composed chiefly of a fibrous protein called **keratin** (KAIR-uh-tin), which is present in all horny growths such as nails, claws, and hoofs. Full-grown human hair is divided into two parts: the hair root and the hair shaft. The **hair root** (HAYR ROOT) is that portion of the hair enclosed within the follicle beneath the skin surface. The **hair shaft** (HAYR SHAFT) is the portion of the hair we see extending above the skin surface.

STRUCTURES OF THE HAIR ROOT

The main structures of the hair root are the follicle, bulb, dermal papilla, sebaceous glands, and arrector pili muscle (**Figure 10-1**).

The hair **follicle** (FAWL-ih-kul) is the tubelike depression or pocket in the skin or scalp that contains the hair root. Hair follicles are distributed all over the body, with the exception of

the palms of the hands and the soles of the feet. The follicle extends downward from the epidermis into the dermis, where it surrounds the dermal papilla. Follicles vary in depth, depending on the thickness and location of the skin. Sometimes, more than one hair will grow from a single follicle. The funnel-shaped mouths of hair follicles make this area breeding places for germs and the accumulation of sebum and dirt. Proper shampooing and rinsing procedures help to minimize the occurrence of scalp and hair disorders caused by the germs and dirt.

The **hair bulb** (HAYR BULB) is club-shaped structure that forms the lower part of the hair root and is the lowest part of the hair strand. The lower part of the hair bulb is hollow and fits over and covers the dermal papilla.

The **dermal papilla** (DUR-mul puh-PIL-uh) is a small, cone-shaped elevation at the base of the hair follicle that fits into the hair bulb. The dermal papilla contains the blood and nerve supply that provides the nutrients needed for hair growth, because of that some people refer to the dermal papilla as the *mother* of the hair.

The sebaceous glands consist of small, saclike structures with ducts that are attached to each hair follicle. They secrete an oily substance called *sebum* that lubricates the skin. Some of the factors that influence sebum production are subject to personal control. These factors are diet, blood circulation, emotional disturbance, stimulation of the endocrine glands, and certain drugs.

- *Diet* influences the general health of the hair. Overindulgence in sweet, starchy, and fatty foods may cause the sebaceous glands to become overactive and to secrete too much sebum, which can cause an oily scalp condition.
- *Blood circulation* is a factor because the hair and scalp derive their nourishment from the blood supplied to them; in turn, proper blood supply depends upon the nutrients derived from the foods eaten.
- *Emotional disturbances* are linked to the health of the hair through the nervous system. The hair's condition is affected by stress.
- *Endocrine glands* are ductless glands. Their secretions go directly into the bloodstream, which in turn influences the welfare of the entire body. The condition of the endocrine glands influences their secretion. During adolescence, endocrine glands are very active; after middle age, their activity usually decreases. Endocrine gland disturbances influence the hair as well as other aspects of health.
- *Medication*, such as hormones, may adversely affect the hair.

The **arrector pili** (ah-REK-tohr PY-leh) is a minute, involuntary muscle fiber in the skin attached to the base of the hair follicle. Strong emotions or cold causes it to contract, which makes the hair stand up straight, resulting in *goose bumps*.

STRUCTURES OF THE HAIR SHAFT

The three main layers of the hair shaft are the cuticle, cortex, and medulla (**Figure 10-2**). The hair **cuticle** (KYOO-tih-kul) is the outermost layer of the hair. It consists of a single overlapping layer of transparent, scalelike cells that look like shingles on a roof.

figure 10-2
Cross section of hair cuticle.

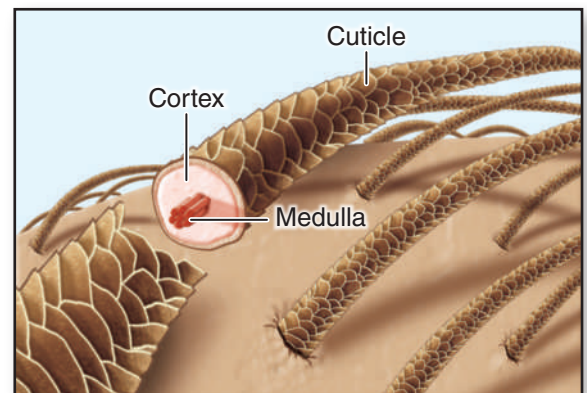
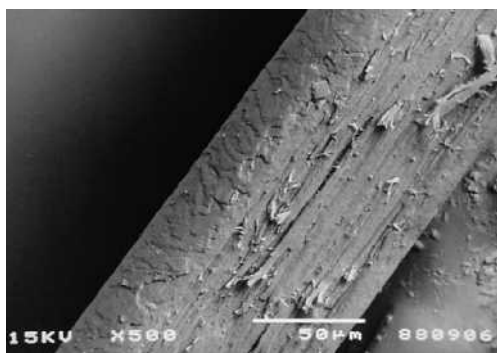


figure 10-3
Hair cuticle layer.



figure 10-4
Hair shaft with part of the hair cuticle stripped off, exposing the cortex.



The cuticle layer provides a barrier that protects the inner structure of the hair as it lies tightly against the cortex. These scales can be felt by holding a strand of hair between the thumb and forefinger of one hand while moving the thumb and fingers of the other hand down the hair strand from its end to the scalp. The hair feels rough when you move from its end to the scalp because you are going against the cuticle layer. A healthy, compact cuticle layer is the hair's primary defense against damage and protects the inner structure of the hair; it is also responsible for the shine and silkiness of the hair. Conversely, heat and certain chemical solutions can raise the scales of the cuticle layer to allow for penetration and absorption of substances into the cortex.

A cross section of hair shows that although the cuticle scales overlap, each individual cuticle scale is attached to the cortex, thereby creating only one cuticle layer (**Figure 10-3**). Swelling the hair with high-pH products, such as oxidation tints, permanent waving solutions, or chemical hair relaxers, raises the cuticle layer and opens the spaces between the scales, allowing liquids to penetrate into the cortex.

The **cortex** (KOR-teks) is the middle layer of the hair. It is a fibrous protein core formed by elongated cells that contain melanin pigment. About 90 percent of the total weight of the hair comes from the cortex. Its unique protein structure provides strength, elasticity, and natural color to the hair. The changes that take place in the hair during chemical services occur within the cortex (**Figure 10-4**).

The **medulla** (muh-DUL-uh) is the innermost layer of the hair and is composed of round cells. Although mature male beard hair contains a medulla, this layer of the hair may be absent in very fine and naturally blond hair.

Photography: Courtesy of P&G Beauty, from the World of Hair by John Gray.

Learn about the Chemical Composition of Hair

After reading this section, you will be able to:

LO 3 Identify and explain the three types of side bonds of the cortex.

Hair is composed of protein that grows from cells that originate within the hair follicle. This is where the hair shaft begins. When these living cells form, they begin a journey upward through the follicle, where they mature through a process called **keratinization** (kair-uh-ti-ni-ZAY-shun). As the newly formed cells mature, they fill up with a fibrous protein called *keratin*, move upward, lose their nuclei, and die. By the time the hair shaft emerges from the scalp, the cells are completely keratinized and no longer living. The hair shaft that we see is a nonliving fiber composed of keratinized protein.

Human hair is approximately 90 percent protein. Protein is made of chemical units called **amino acids** (uh-MEE-noh AS-udz) and amino acids are made of elements. The essential elements in the human hair are carbon, oxygen, hydrogen, nitrogen, and sulfur, often referred to as the *COHNS elements* (KOH-nz EL-uh-mentz). These elements are also found in skin and nails. **Table 10-1** shows the percentage of each element in a typical strand of hair; this chemical composition, however, varies with color.

table 10-1

THE COHNS ELEMENTS

Element	Percentage in Normal Hair
Carbon	51
Oxygen	21
Hydrogen	6
Nitrogen	17
Sulfur	5

Proteins are made of long chains of amino acids. These amino acids are joined end to end in a definite order by chemical bonds known as **peptide bonds** (PEP-tyd BAHNDZ) or **end bonds** (END BAHNDZ). A long chain of amino acids linked by peptide bonds is called a **polypeptide chain** (pahl-ee-PEP-tyd CHAYN). **Proteins** (PROH-teenz) are long, coiled complex polypeptides made of amino acids. The spiral shape of a coiled protein is called a **helix** (HEE-licks), which is created when the polypeptide chains intertwine with each other (Figure 10-5).

SIDE BONDS OF THE HAIR CORTEX

Within the hair cortex, a more complex structure is formed when millions of polypeptide chains are cross-linked by **side bonds** (SYD BAHNDZ) (formerly known as *cross-bonds*), to form a ladderlike structure (Figure 10-6). Side bonds are of three types, consisting of hydrogen, salt, and disulfide bonds, which account for the strength and elasticity of human hair. These bonds play a critical role in the blowdrying, wet setting, thermal styling, and chemical processes. Table 10-2 summarizes the properties of the types of bonds found within the protein structure of hair.

table 10-2

BONDS OF THE HAIR

Bond	Type	Strength	Broken By	Re-formed By
Hydrogen	Side bond	Weak, physical	Water or heat	Drying or cooling
Salt	Side bond	Weak, physical	Changes in pH	Normalizing pH
Disulfide	Side bond	Strong, chemical	Thio perms and thio relaxers	Oxidation with neutralizer
			Hydroxide relaxers	Cannot be re-formed; water converts broken disulfide bonds to lanthionine bonds
			Extreme heat	Cannot be re-formed; water converts broken disulfide bonds to lanthionine bonds
Peptide	End bond	Strong, chemical	Chemical depilatories or cutting	Cannot be re-formed; hair dissolves

figure 10-5

Polypeptide chains intertwine in a spiral shape called a **helix**.

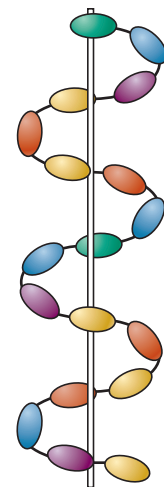
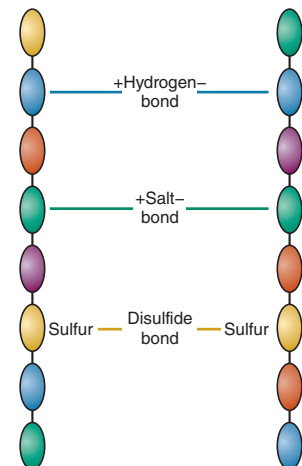


figure 10-6

Side bonds between polypeptide chains.



A **hydrogen bond** (HY-druh-jun BAHND) is a weak, physical, cross-link side bond that is easily broken by water or heat. Although individual hydrogen bonds are weak, they are so numerous in the hair that they account for about one-third of the hair's overall strength. Hydrogen bonds are broken when hair gets wet (**Figure 10-7**), which allows the hair to be stretched. The hydrogen bonds re-form when the hair dries, which explains how blowdrying techniques work when the barber works with a comb or brush to style the hair into place while drying.

A **salt bond** (SAWLT BAHND) is a weak, physical side bond, which cross-links the polypeptide chains, but reacts to changes in pH. Salt bonds depend on pH and are easily broken by strong acidic or alkaline solutions (**Figure 10-8**). Although weak, salt bonds are numerous and account for another one-third of the hair's total strength.

A **disulfide bond** (dy-SUL-fyed BAHND), also known as a *sulfur bond*, is a strong covalent bond that is different from the physical side bond of hydrogen or salt bonds. Disulfide bonds join the sulfur atoms of two neighboring **cysteine** (SIS-tuh-een) amino acids to create one *cystine* (SIS-teen). The cystine joins together two polypeptide strands. Although there are fewer disulfide bonds in the hair, they are stronger than hydrogen or salt bonds and account for the final third of the hair's total strength. Unlike hydrogen and salt bonds, disulfide bonds are not broken by heat or water—they are broken by permanent waves and chemical relaxers that alter the shape of hair. Normal amounts of heat, such as those used for normal thermal styling, do not break disulfide bonds. However, extremely high heat, such as that from a very hot flat iron, can break the disulfide bond. Thioglycolate permanent waves break disulfide bonds, which are then

figure 10-7
Changes in hair cortex during wet setting.

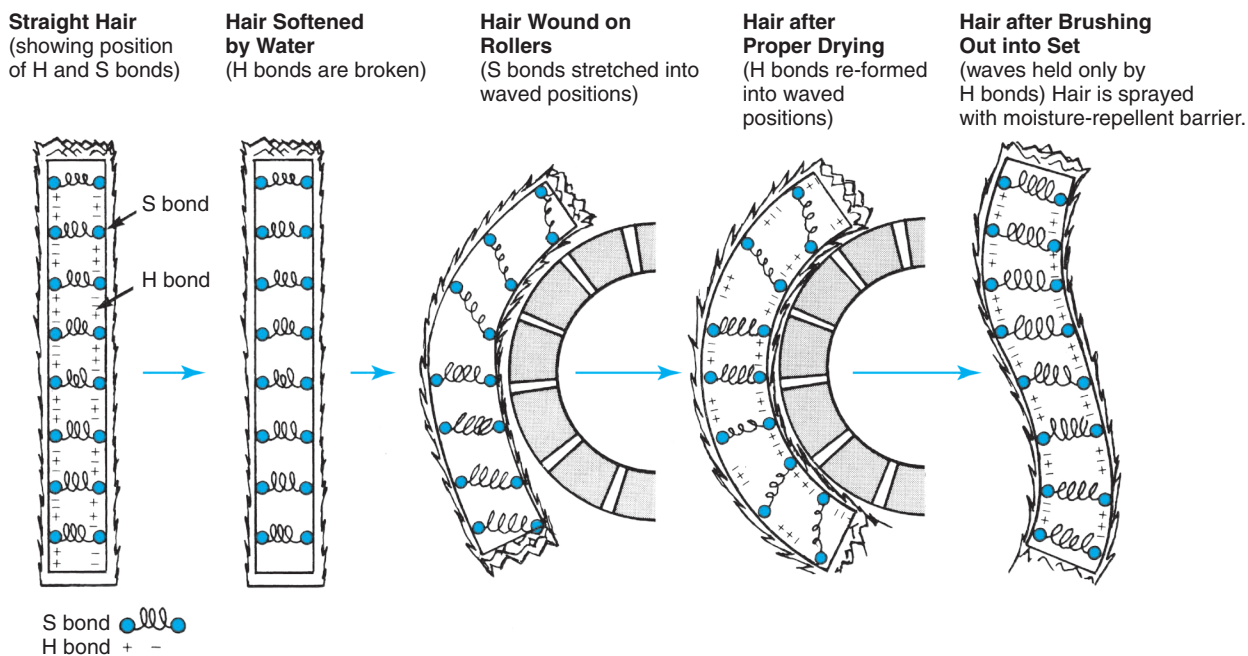
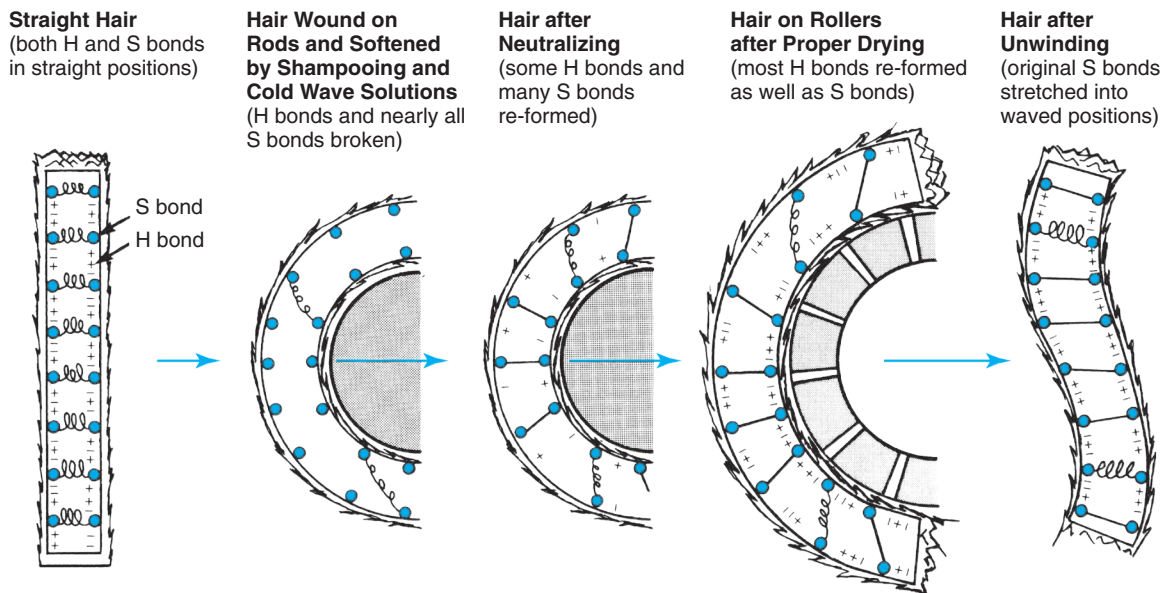


figure 10-8

Changes in hair cortex during permanent waving.



re-formed with neutralizers. Hydroxide chemical hair relaxers also break disulfide bonds, which are then converted to lanthionine bonds when the relaxer is rinsed from the hair. Disulfide bonds broken by hydroxide relaxers are permanently broken and cannot be re-formed. The more disulfide bonds in the hair, the more resistant it will be to chemical processes.

HAIR PIGMENT

Natural hair color is the result of the melanin pigment found within the cortex.

Melanin are the tiny grains of pigment in the cortex that give natural color to the hair. There are two types of melanin:

- **Eumelanin** (yoo-MEL-uh-nin) provides brown and black color to hair.
- **Pheomelanin** (fee-oh-MEL-uh-nin) provides natural hair colors that range from red and ginger to yellow and light blond tones.

All natural color is dependent on the ratio of eumelanin to pheomelanin, along with the total number and size of the pigment granules. Gray hair contains only a few scattered melanin granules and white hair contains no melanin.

WAVE PATTERN

The **wave pattern** (WAYV PAT-URN) of hair refers to the shape of the hair strand and is described as straight, wavy, curly, and extremely curly (**Figure 10-9**).

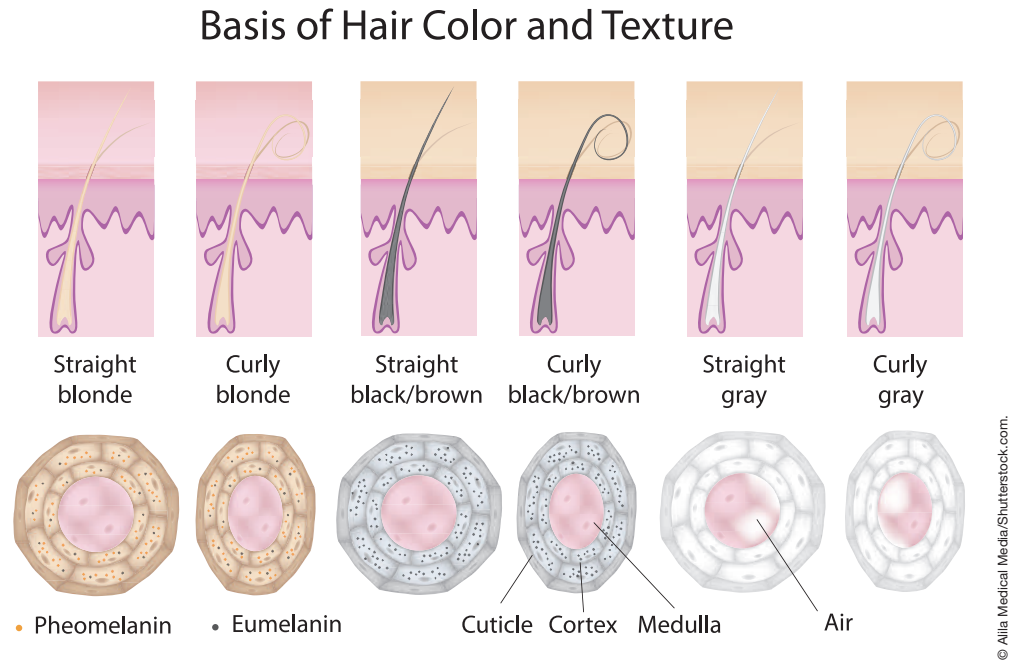
Natural wave patterns are the result of genetics. Although there are many exceptions, as a general rule, Asians and Native Americans tend to have extremely straight hair, Caucasians tend to have straight, wavy, or curly hair, and African Americans tend to have extremely curly hair.

figure 10-9

Straight, wavy, curly, and extremely curly hair strands.



figure 10-10
Individuals with curly hair often have straighter hair in the crown and tighter curls in other areas.



DID YOU KNOW? The term *hair color* (two words) refers to the color of hair created by nature. *Haircolor* (one word) is the term used in the beauty industry to refer to artificial haircoloring products. Gray hair is caused by the absence of melanin. Gray hair grows from the hair bulb in exactly the same way that pigmented hair grows. It has the same structure, but without the melanin pigment.

But straight, wavy, curly, and extremely curly hairs occur in all races—anyone of any race, or mixed race, can have hair with varying degrees of curl from straight to extremely curly. The wave pattern may also vary from strand to strand on the same person’s head. It is not uncommon for an individual to have different amounts of curl in different areas of the head. Individuals with curly hair often have straighter hair in the crown and tighter curls in other areas (**Figure 10-10**).

Several theories attempt to explain the cause of natural curly hair, but there is no single, definite answer that explains why some hair grows straight and other hair grows curly. The most popular theory claims that the shape of the hair’s cross section determines the amount of curl. This theory claims that hair with a round cross section is straight, hair with an oval-to-flattened oval cross section is wavy or curly, and hair with a flattened-to-flattened oval cross section is extremely curly (**Table 10-3**).

table 10-3
WAVE PATTERN AND CROSS SECTIONS

	Wave Pattern	Shape of Cross Section
	Straight Hair	Round cross section
	Wavy or Curly Hair	Oval to round cross section
	Extremely Curly Hair	Elliptical cross section

EXTREMELY CURLY HAIR

Extremely curly hair grows in long twisted spirals. Cross sections appear flattened and vary in shape and thickness along their length. Compared to straight or wavy hair, which tends to possess a fairly regular and uniform diameter along a single strand, extremely curly hair is of fairly irregular thickness, showing varying diameters along a single strand. Some extremely curly hair have a natural tendency to form a coil like a telephone cord. Coiled hair usually has a fine texture, with many individual strands winding together to form the coiled locks. Extremely curly hair often has low elasticity, breaks easily, and has a tendency to knot, especially on the ends. Gentle scalp manipulations, conditioning shampoo, and a detangling rinse help minimize tangles.

Discuss Hair Growth

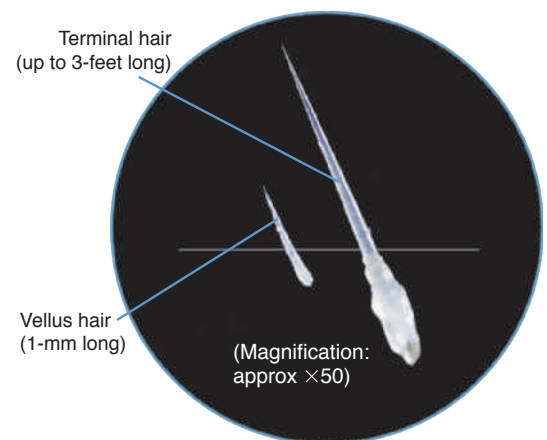
After reading this section, you will be able to:

LO 4 Name and describe the three phases of the hair growth cycle.

The two main types of hair are vellus hair and terminal hair (Figure 10-11).

- **Vellus hair** (VEL-us HAYR), or **lanugo** (luh-NOO-goh) hair, is the short, fine, unpigmented, downy hair. It almost never has a medulla or melanin and helps in the efficient evaporation of perspiration. In adults, vellus hair is found in places normally considered hairless (forehead, eyelids, and bald scalp).
- **Terminal hair** (TUR-mih-nul HAYR) is the long, coarse hair found on the scalp, legs, arms, and bodies of males and females. Hormonal changes during puberty cause some areas of fine vellus hair to be replaced with thicker terminal hair. All hair follicles are capable of producing either vellus or terminal hair, depending on genetics, age, and hormones. In occurrences of male pattern baldness, the follicles stop making terminal hair and revert back to producing the vellus type.

figure 10-11
Vellus hair and terminal hair.



GROWTH CYCLES OF HAIR

The average growth rate of healthy hair on the scalp is about $\frac{1}{2}$ (0.5) inch per month. The rate of growth will differ over specific parts of the body, between sexes, among races, and with age. The growth of scalp hair occurs more rapidly between the ages of 15 and 30 and declines sharply between 50 and 60. Hair growth is also influenced by such factors as the seasons of the year, nutrition, and hormonal changes within the body.

In normal, healthy hair, hair growth occurs in cycles. Each complete cycle has three phases that are repeated over and over throughout life. These three phases are known as the *anagen*, *catagen*, and *telogen* phases.

Anagen: The Growth Phase

During the **anagen phase** (AN-uh-jen FAYZ), or *growth phase*, new hair is produced. New cells are manufactured in the hair follicle. During this phase, hair cells are produced faster than any other normal cell in the body. About 90 percent of scalp hair is in the anagen phase—that is, growing—at any given time. The anagen phase generally lasts from 3 to 5 years, but can last as long as 10 years. The anagen phase determines how long the hair will grow before shedding; therefore, the longer the anagen phase is, the longer the hair can grow. This explains why some people can grow their hair to extremely long lengths and others cannot.

Catagen: The Transition Phase

The **catagen phase** (KAT-uh-jen FAYZ) is a transition period between the growth and resting phases of a hair strand. During the catagen phase the follicle shrinks, the hair bulb disappears, and the shrunken root end forms a rounded club. The catagen phase lasts from 1 to 2 weeks. Less than 1 percent of the hair is in the catagen phase at any given time.

Telogen: The Resting Phase

The **telogen phase** (TEL-uh-jen FAYZ), or *resting phase*, is the final phase of the hair growth cycle. The hair either sheds during this phase or remains in place until it is pushed out by the growth of a new hair in the next anagen phase. Less than 10 percent of scalp hair is in the telogen phase at any given time. The telogen phase lasts for approximately 3 to 6 months or until the hair is shed. As soon as the telogen phase ends, new hair in the anagen phase begins the cycle again. On average, the entire growth process repeats itself once every 4 or 5 years.

Hair growth is not increased by shaving, trimming, cutting, singeing, or the application of ointments or oils. Hair growth is also not stimulated by scalp massage, although it does increase blood flow and relax the nerves of the scalp. Minoxidil and finasteride are the only treatments that have been scientifically proven to increase hair growth and are approved for that purpose by the Food and Drug Administration (FDA). Products that claim to increase hair growth are regulated as drugs and are not cosmetics.

NORMAL HAIR SHEDDING

It is normal to lose an average of 75 to 100 hairs per day. This shedding process makes room for new hair that is in the process of growing. Hair loss beyond this estimated average indicates some problem. Eyebrow hairs and eyelashes are replaced every 4 to 5 months.

GROWTH PATTERNS

It is important when cutting and styling hair to consider the hair's natural growth patterns. Doing so will produce a more natural-looking haircut and a style that the client will have less trouble duplicating for everyday wear. Growth patterns that result in creating hair streams, whorls, and cowlicks are determined by the angle at which the hair emerges from the hair follicle.

- A **hair stream** (HAYR STREEM) is hair that flows in the same direction. It is the result of follicles being arranged and sloping in a uniform manner. When two such streams slope in opposite directions, they form a natural part in the hair.
- A **whorl** (WHORL) is hair that grows in a circular or swirl pattern. Whorls are most often seen at the crown.
- A **cowlick** (KOW-lik) is a tuft of hair that stands straight up. Cowlicks are usually more noticeable at the front hairline, but they may be located anywhere on the scalp. Styles should be chosen to minimize the upright effects of cowlicks.

MYTHS AND FACTS ABOUT HAIR GROWTH

As a barber, you may hear opinions about hair growth from your clients or from other professionals. Here are some myths and facts about hair growth:

Myth. Shaving, clipping, and cutting the hair on the head makes it grow back faster, darker, and coarser.

Fact. Although it may *seem* to grow back faster, darker, and coarser, shaving or cutting the hair on the head has no effect on hair growth. When hair is blunt cut to the same length, it may grow back more evenly.

Myth. Scalp massage increases hair growth.

Fact. Scalp massages are very stimulating to the scalp and can increase blood circulation, relax the nerves in the scalp, and tighten the scalp muscles. However, it has not been scientifically proven that any type of stimulation or scalp massage increases hair growth.

Myth. Gray hair is coarser and more resistant than pigmented hair.

Fact. Other than the lack of pigment, gray hair is exactly the same as pigmented hair. Although gray hair may be resistant, it is not resistant simply because it is gray. Pigmented hair on the same person's head is just as resistant as the gray hair. Gray hair is simply more noticeable than pigmented hair.

Myth. The amount of natural curl is always determined by racial background.

Fact. Anyone of any race, or mixed race, can have hair from straight to extremely curly. It is also true that within races, individuals have hair with varying degrees of curl in different areas of the head.

Understand Hair Loss

After reading this section, you will be able to:

LO5 Identify and define seven types of hair loss.

LO6 Identify and describe two FDA-approved treatments for hair loss.

We all lose some hair every day as a result of the normal shedding that takes place during the hair's natural growth and replacement cycle. A hair that is shed in the telogen phase is replaced by a new hair, in that same follicle, in

the next anagen phase. This natural shedding of hair accounts for normal daily hair loss, which averages 75 to 100 hairs per day.

THE EMOTIONAL IMPACT OF HAIR LOSS

Although the medical community does not always recognize hair loss as a medical condition, the anguish felt by those who suffer abnormal hair loss is very real and often overlooked. Results from a study that investigated perceptions of bald and balding men showed that compared to men with hair, bald men were perceived as

- less physically attractive (by both sexes);
- less assertive;
- less successful;
- less personally likeable;
- older than their actual age (average 5 years).

Over 63 million men and women in the United States suffer from abnormal hair loss. As a professional barber, it is likely that you will be the first person that a hair loss sufferer confides in, so it is important that you have a basic understanding of the different types of hair loss and the services that are available.

IDENTIFY TYPES OF ABNORMAL HAIR LOSS

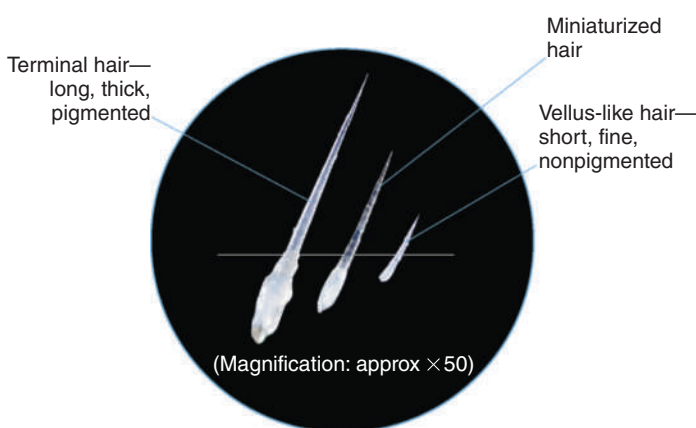
Abnormal hair loss is called **alopecia** (al-oh-PEE-shah). Understandably, hair loss for either sex can be a traumatic experience, so conversations about it should be handled tactfully. Alopecia may appear in different forms as a result of a variety of abnormal conditions. These forms include androgenic alopecia, alopecia prematura, alopecia areata, alopecia totalis, alopecia universalis, alopecia senilis, and alopecia syphilitica.

Androgenic alopecia (an-druh-JEN-ik al-oh-PEE-shah) is hair loss that occurs as a result of genetics, age, and hormonal changes that cause the miniaturization of terminal hair, converting it to vellus hair (**Figure 10-12**).

Androgenic alopecia can begin as early as in the teens (**alopecia prematura** ([al-oh-PEE-shah pree-muh-TOO-ruh]) and is frequently seen by the age of 40. Almost 40 percent of men and women show some degree of hair loss by age 35. In men, androgenic alopecia is known as *male pattern baldness* and usually progresses to the familiar horseshoe-shaped fringe of hair (**Figure 10-13**).

Alopecia areata (al-oh-PEE-shah ay-reh-AH-tuh) is an autoimmune disorder that causes the affected hair follicles to be mistakenly attacked by the person's own immune system, which stops hair growth at the anagen phase. It is characterized by the sudden falling out of hair in round patches, which creates bald spots, and can progress to total scalp hair loss,

figure 10-12
Miniaturization of the hair follicle.



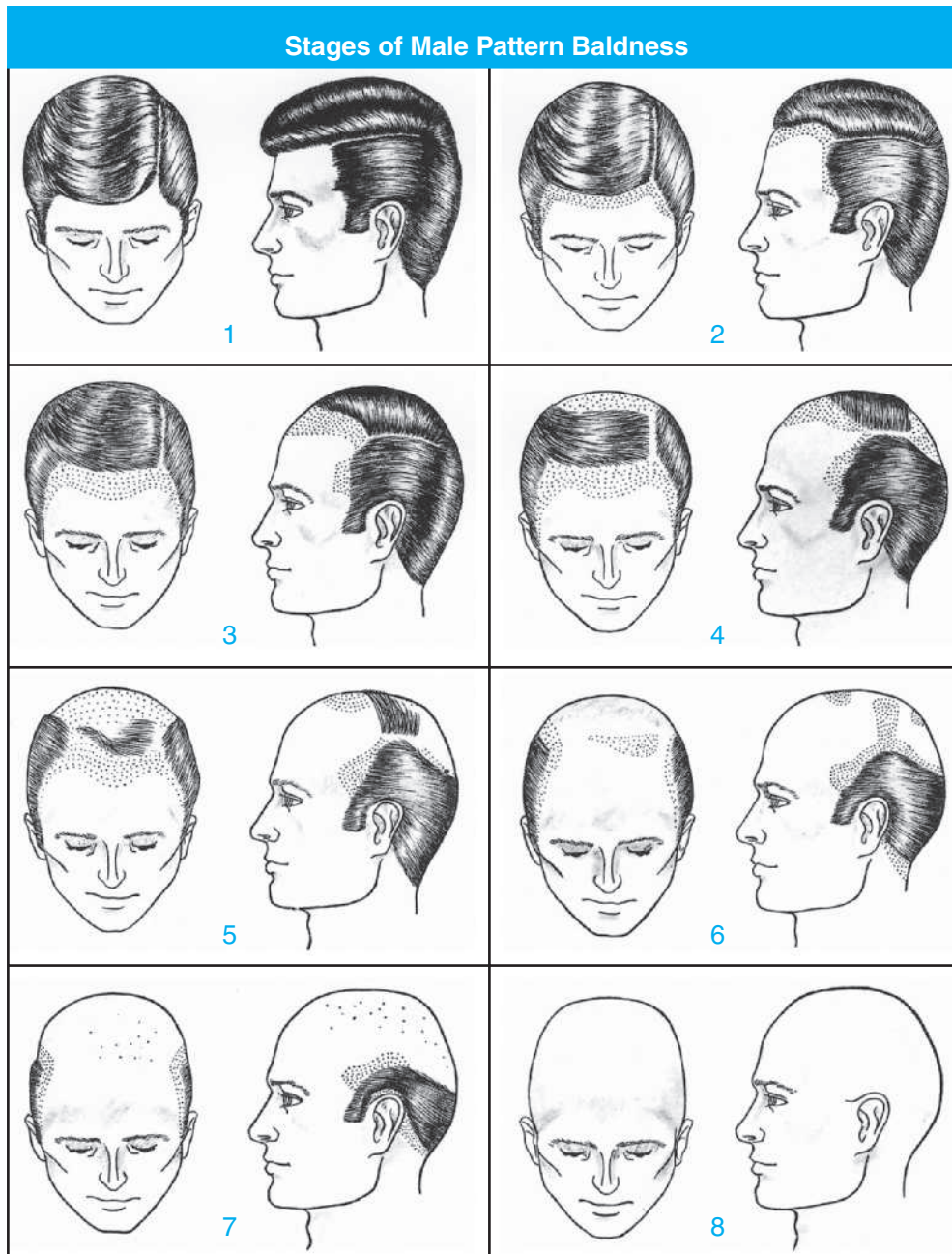


figure 10-13
Gradual balding process.

known as **alopecia totalis** (al-oh-PEE-shah toe-TAL-is), or complete body hair loss, called **alopecia universalis** (al-oh-PEE-shah yoo-nih-vur-SAA-lis). It is a highly unpredictable skin disease that may occur on the scalp and elsewhere on the body. Clients who exhibit symptoms of alopecia areata should be referred to a physician. **Alopecia senilis** (al-oh-PEE-shah seh-NIL-is) is the normal loss of scalp hair occurring in old age and is permanent. **Alopecia syphilitica** (al-oh-PEE-shah sif-il-IT-ih-kuh) is caused by syphilis. The non-inflamed bald areas look molted or moth-eaten, and the disease may also affect the beard and eyebrow areas. The hair usually grows back once the disease is cured.

DESCRIBE HAIR LOSS TREATMENTS

The only FDA-approved hair loss treatments, which have been proven to stimulate hair growth, are minoxidil and finasteride.

- *Minoxidil* is a topical treatment that has been proven to stimulate hair growth. Applied to the scalp twice a day, it is sold over the counter as a nonprescription drug under the brand names Rogaine and Theroxidil. Minoxidil is available for both men and women. It is available in two strengths: 2 percent (regular) and 5 percent (extra strength). It is not known to have any negative side effects.
- *Finasteride* is an oral prescription medication for men, sold under the brand name Propecia, and is for use only by men. This drug is extremely effective in slowing the rate of hair loss and can even grow new hair; however, once you stop using the drug all the hair loss returns. Although finasteride is considered more effective and convenient than minoxidil, side effects are possible, which include weight gain and loss of sexual function.

In addition to the medicinal treatments described above, there are also several surgical options available. A hair transplant is the most common *permanent* hair replacement technique. The process consists of removing small sections of hair that include the follicle, papilla, and bulb from areas of thick hair growth and transplanting them into the bald area. Only licensed surgeons may perform this procedure and several surgeries are usually necessary to achieve the desired results. The cost of each surgery ranges from about \$8,000 to over \$20,000.

With proper training, barbers can offer several nonmedical options to camouflage hair loss. These include hair replacement systems, wigs, hair weaves, and hair extensions (see Chapter 15).

Recognize Disorders of the Hair and Scalp

After reading this section, you will be able to:

- LO 7 Identify and define common hair disorders.
- LO 8 Define common scalp disorders and identify those requiring medical attention.

RECOGNIZE DISORDERS OF THE HAIR

Hair disorders are noncontagious conditions that include disorders such as canities, hypertrichosis, trichoptilosis, trichorrhexis nodosa, monilethrix, and fragilitas crinium.

- **Canities** (kah-NISH-ee-eez) is the technical term for gray hair. Congenital canities exists at or before birth and generally is part of a disorder known as *albinism* (albino) that affects all pigment throughout the body. Acquired canities develops with age and is the result of genetics.

- Ringed hair is a variation of canities, characterized by alternating bands of gray and pigmented hair throughout the length of the hair strand.
- **Hypertrichosis** (hy-pur-trih-KOH-sis), also known as *hirsuties* (hur-SOO-teez) is a condition of abnormal growth of hair. It is characterized by the growth of terminal hair on areas of the body that are normally covered with only vellus hair.
- **Trichoptilosis** (trih-kahp-tih-LOH-sus) is the technical term for split ends. Hair conditioning and oils will smooth split ends, but the only way to remove them is to cut them.
- **Trichorrhexis nodosa** (trik-uh-REK-sis nuh-DOH-suh) is the technical term for knotted hair. It is characterized by brittleness and nodular swellings along the hair shaft. Treatments include softening the hair with conditioners and moisturizers.
- **Monilethrix** (mah-NIL-ee-thriks) is the technical term for beaded hair. The hair breaks easily between the beads or nodes. Treatments include hair and scalp conditioning.
- **Fragilitas crinium** (fruh-JIL-ih-tus KRY-nee-um) is the technical term for brittle hair. The hairs may split at any point along their length. Treatments include hair and scalp conditioning and haircutting above the split to prevent further damage.

IDENTIFY DISORDERS OF THE SCALP

The skin of the scalp is constantly renewing itself, just as skin does on all parts of the body. Problems with the skin of the scalp can range from irritating and embarrassing to painful and dangerous.

RECOGNIZE TYPES OF DANDRUFF

Pityriasis (pit-ih-RY-uh-sus) is the technical term for dandruff, which is characterized by the excessive production and accumulation of skin cells. Instead of the normal, one-at-a-time shedding of tiny individual skin cells, dandruff is the shedding of an accumulation of large, visible clumps of skin cells. It can be easily mistaken for dry scalp because the symptoms of both conditions are a flaky, itchy, irritated scalp, but a dry scalp does not have the oily scalp that is common to dandruff. An unhealthy scalp—due to poor blood circulation, infection, injury, improper diet, or poor personal hygiene—can contribute to this accumulation, as can the use of strong shampoos and/or insufficient rinsing of the hair. Although the cause of dandruff has been debated for over 150 years, current research confirms that dandruff is the result of a fungus called **malassezia** (mal-uh-SEEZ-ee-uh), formerly named *pityrosporum*. Malassezia is a naturally occurring fungus that is present on all human skin, but develops the symptoms of dandruff when it grows out of control. Factors including stress, age, hormones, and hygiene can cause the fungus to multiply and dandruff symptoms to worsen. Antidandruff shampoos containing antifungal ingredients such as pyrithione zinc, selenium sulfide, or ketoconazole help control dandruff by suppressing the growth of malassezia.

figure 10-14
Pityriasis capitis simplex, commonly known as dandruff.



figure 10-15
Pityriasis steatoides.

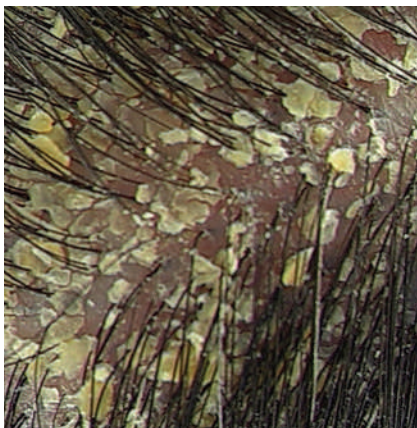


figure 10-16
Tinea capitis.



The two principal types of dandruff are pityriasis capitis simplex and pityriasis steatoides.

- **Pityriasis capitis simplex** (pit-ih-RY-uh-sus KAP-ih-tus SIM-pleks) is the technical term for classic dandruff, characterized by scalp irritation, large flakes, and an itchy scalp (**Figure 10-14**). The scales may be attached to the scalp in masses or scattered loosely throughout the hair. Topical treatments for controlling dandruff include the use of antidandruff shampoos, scalp massage and treatments, and medicated scalp ointments.
- **Pityriasis steatoides** (pit-ih-RY-uh-sus stee-uh-TOY-deez) is a more severe form of dandruff that is characterized by an accumulation of greasy or waxy scales mixed with sebum (**Figure 10-15**). This excessive shedding mixed with sebum causes the scales to adhere to the scalp in patches, where they can cause itching and irritation. When this condition is accompanied by redness and inflammation, it is called *seborrheic dermatitis*, which can also be found on the eyebrows and beard. You should not perform a service on anyone who has dandruff, as the scalp is irritated and itchy. Antidandruff shampoos can be recommended to a client with mild conditions, but severe cases should be referred to a physician.

At one time, dandruff was thought to be contagious; however, current research has determined that it is not. Regardless of whether or not dandruff is contagious, barbers are prohibited by state barber board rules to use their tools and implements on more than one client without first disinfecting the items. Barbers must always practice approved cleaning and disinfection procedures in the barbershop before and after each client service. It is also advisable to wear gloves when shampooing a client with a dandruff condition.

IDENTIFY FUNGAL INFECTIONS: TINEA

Tinea (TIN-ee-uh) is the medical term for ringworm. Ringworm is caused by fungal organisms and is characterized by itching, scales, and, sometimes, painful circular lesions. It usually starts with a small, reddened patch of little blisters that spreads outward and then heals in the middle, with a scalelike appearance. If the ringworm has spread, several patches may be present at one time.

All forms of tinea are highly contagious. Infected skin scales or hair containing the fungi are known to spread the disease; public showers, swimming pools, and unclean personal articles (such as brushes) are also sources of transmission. Practicing approved cleaning and disinfection procedures will help prevent the spread of ringworm in the barbershop. Clients with suspected or confirmed tinea should be referred to a physician for medical treatment and must not receive services in the barbershop until the condition has cleared.

- **Tinea capitis** (TIN-ee-uh KAP-ih-tus) is commonly known as *ringworm* of the scalp (**Figure 10-16**). It is characterized by red papules or spots at the openings of the hair follicles. As the patches spread, the hair becomes brittle and lifeless and breaks off, leaving a stump, or falls from the enlarged, open follicles.

Photography: Courtesy of P&G Beauty.

Photography: Courtesy of P&G Beauty.

Courtesy of Robert A. Silverman, M.D., Clinical Associate Professor, Department of Pediatrics, Georgetown University.

- **Tinea barbae** (TIN-ee-uh BAR-bee) is a superficial fungal infection caused by dermatophytes occurring chiefly over the bearded area of the face (Figure 10-17). Beginning as small, round, slightly scaly, inflamed patches, the areas enlarge, clearing up somewhat at the center with elevation at the borders. As the parasites invade the hairs and follicles, hard, lumpy swellings develop. In severe cases, pustules form around the hair follicles and rupture, forming crusts. In the later stage, the hairs become dry, break off, and fall out or are readily extracted. Tinea barbae is highly contagious and medical treatment is required.
- **Tinea favosa** (TIN-ee-uh fah-VOH-suh) is also known as *tinea favus* (TIN-ee-uh FAY-vus) or *honeycomb ringworm*. It is characterized by dry, sulfur-yellow, cuplike crusts on the scalp having a peculiar, musty odor—these are called **scutula** (SKUT-yoo-lah). Scars from favus are bald patches that are pink or white and shiny. Tinea favosa is very contagious and should be referred to a physician.

figure 10-17
Folliculitis barbae or tinea (ringworm) barbae.



RECOGNIZE PARASITIC INFESTATIONS

The two most common parasitic infestations barbers may see in the barbershop are pediculosis capitis and scabies.

- **Pediculosis capitis** (puh-dik-yuh-LOH-sis KAP-ih-tus) is the infestation of the hair and scalp with head lice (Figure 10-18). The parasites feed on the scalp and cause severe itching. You can distinguish head lice from dandruff or other conditions by looking at the scalp and hair closely—with a magnifying glass if necessary. The scalp usually appears reddened from scratching, and you should be able to see mature lice moving through the hair. The nits, or eggs, attach to individual hair strands and are difficult to remove without a special comb made for that purpose. The nits can also withstand some products used to treat the infestation, so repeat applications of the preparation is usually required. Commercially prepared products for the treatment of head lice are sold over the counter and are readily available at most drug stores.

The head louse is transmitted from one person to another by contact with infested hats, combs, brushes, and other personal items; so following proper cleaning and disinfection procedures is extremely important.

Head lice are tenacious creatures that can live away from the human body for up to 48 hours, so it is equally important to disinfect all household and personal items to avoid reinfestation. Occurrences of head lice must not be treated in the barbershop, and the client should be referred to a physician or pharmacist.

- **Scabies** is a highly contagious skin disease caused by mites that burrow under the skin. Vesicles and pustules usually form from the irritation caused by the parasites or from scratching the affected areas (Figure 10-19). Excessive itching is a characteristic of scabies and can lead to infections of the affected area. A client with this condition should be referred to a physician. As with all disorders and diseases, the practice of approved cleaning and disinfection procedures will help to limit the spread of scabies.

figure 10-18
Head lice.



figure 10-19
Scabies.



RECOGNIZE BACTERIAL INFECTIONS

Barbers may also encounter bacterial staphylococci infections in the barbershop in the form of different types of folliculitis. Folliculitis can occur anywhere on the body as a result of bacterial or viral infection and is characterized by the inflammation or infection of one or more hair follicles. Some of its more common forms are folliculitis barbae, pseudofolliculitis barbae, sycosis vulgaris, furuncles, and carbuncles.

Folliculitis barbae and *pseudofolliculitis barbae* are inflammations of the follicle caused by bacterial or viral infection, irritation, or ingrown hairs. The cause of the inflammation is what differentiates the two conditions, but both can be triggered from damaged follicles caused by friction, blockages, or improper shaving methods, for example, close shaving, too much pressure, dull blades, dirty blades, or shaving against the grain.

Folliculitis barbae (fah-lik-yuh-LY-tis BAR-bee), or barber's itch, is an infection of the hair follicles characterized by inflamed pustules in the bearded areas of the face and neck that may have hairs growing through the pustule.

- Staphylococcus bacteria most often cause the infection; however, a herpes simplex virus can spread to other hair follicles during shaving, resulting in similar pustule formation and infection.
- Treatments include topical or oral antibiotics as prescribed by a physician.

Pseudofolliculitis barbae (SUE-doe-fah-lik-yuh-LY-tis BAR-bee), also referred to as *razor bumps*, is a chronic inflammatory condition that resembles folliculitis in terms of papules and pustules, but is generally accepted to be caused by ingrown hair. The disorder is most often seen in men and women of Mediterranean, African, Hispanic, and Jewish ethnicity, but can occur to anyone with any type of hair texture.

- Pseudofolliculitis barbae can be caused by improper shaving or by broken hair below the skin surface that grows into the side of the follicle, causing irritation and swelling that cuts off oxygen to the bottom of the follicle. Bacteria then have a perfect environment for growth, which can lead to the development of pus and, if left untreated, folliculitis.
- Nonmedical treatments for pseudofolliculitis barbae include the use of proper shaving techniques, antibiotic or antifungal preparations, and the use of electric razors or razors designed to cut hair above the skin to minimize outbreaks and irritation. Clients should also be instructed to change razor blades frequently as dull, dirty blades also increase all types of folliculitis. Warm compresses and clear gel masks that soothe and heal may help to relieve symptoms.

With an understanding of folliculitis and pseudofolliculitis, barbers can help clients to determine if the condition is caused by chemical or mechanical means. Preparations that contain salicylic acid to break up impactions and kill bacteria are available for the prevention of ingrown hairs. Physicians may prescribe topical and oral antibiotics for more serious conditions. When mechanical causes, such as improper shaving, are the source of the condition, the barber can offer the client some instructions in proper shaving techniques.

Sycosis vulgaris (sy-KOH-sis vul-GAYR-is), also known as *sycosis barbae*, is a chronic bacterial infection involving the areas surrounding the follicles of

the beard and mustache areas. It may be contracted through the use of contaminated towels or implements or by contact with public resting areas. The condition can be worsened by irritation caused by shaving or continual nasal discharge. The main lesions are papules and pustules pierced by hairs and crusts that form after eruption. The surrounding skin is tender, reddened, swollen, and itchy. Medical treatment is required and a client with this condition should be referred to a physician. This infection should not be confused with **tinea sycosis** (TIN-ee-uh sy-KOH-sis), which is due to ringworm fungus.

A **furuncle** (FYOO-rung-kul), or boil, is an acute bacterial infection of a hair follicle, producing constant pain. A furuncle is the result of an active inflammatory process limited to a definite area that subsequently produces a pustule perforated by a hair (**Figure 10-20**). A client with this condition should be referred to a physician.

A **carbuncle** (KAHR-bung-kul) is the result of an acute, deep-seated bacterial infection in the subcutaneous tissue. It appears similar to a furuncle, but is larger and should also be referred to a physician.

Learn How to Perform a Hair and Scalp Analysis

After reading this section, you will be able to:

- LO 9** Identify the factors to be observed and considered during a hair and scalp analysis.

Barbering services include a variety of applications that benefit from a barber's ability to analyze the condition of a client's scalp and hair. In addition to wave and growth patterns, barbers should be able to analyze the hair's texture, density, porosity, and elasticity.

Knowledge and skill in performing a hair analysis can be acquired by observation and practice using the senses of sight, hearing, smell, and touch.

- **Sight.** Observation will impart some knowledge immediately, such as whether the hair looks dry or oily. Sight alone, however, will not provide an accurate judgment of the hair's quality.
- **Hearing.** Some clients will volunteer information about their hair, health problems, or experiences with products and medications. Since all of these factors are important when deciding how to treat the hair, it is advisable to listen carefully.
- **Smell.** Certain scalp disorders will create an odor. If the client is in general good health and the scalp is clean, the hair should be odor free.
- **Touch.** The sense of touch is key to analyzing hair condition and texture. This sense needs to be developed to its fullest capacity for the barber to provide truly professional services.

PERFORM A SCALP ANALYSIS

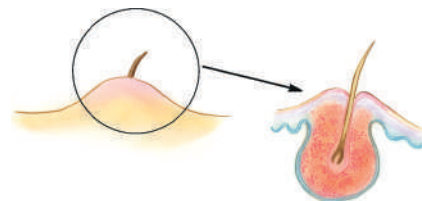
Always begin the analysis with checking the scalp first since any disorders or conditions found there could prohibit a service from being performed.



DID YOU KNOW?

The term *barber's itch* has been used interchangeably in the past to describe many skin conditions, from folliculitis to tinea sycosis to razor burn. Currently, the term is applied most frequently to a folliculitis barbae condition.

figure 10-20
Furuncle (boil).



When analyzing the scalp, comb through the hair gently while parting it to check for hypertrophies, abrasions, and parasites; do not scrape the scalp.

- Do not begin any service if parasites are present.
- Do not proceed with chemical services if there are signs of irritation or abrasions.
- Hypertrophies, such as moles or warts, require extreme care when combing through the hair to avoid catching the comb on these abnormal skin growths. If you see changes in a hypertrophy from one client visit to another, suggest that your client see a physician. Barbers play a significant role in the early detection of many skin cancers, so any unusual-looking pigmentation or growth on the scalp should be referred to a physician.

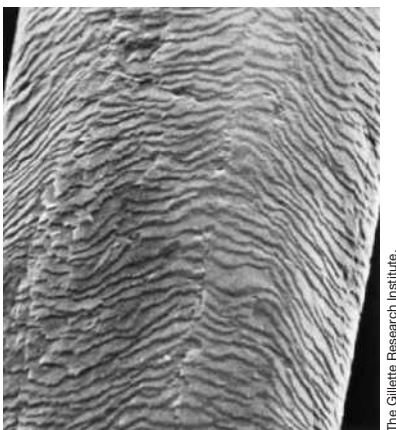
PERFORM A HAIR ANALYSIS

Observe and consider hair texture, density, porosity, and elasticity when performing a hair analysis.

Hair texture (HAYR TEKS-chur) refers to the degree of coarseness or fineness of individual hair strands, which may vary on different parts of the head. Hair texture is measured by the diameter of the hair strand and is classified as coarse, medium, or fine.

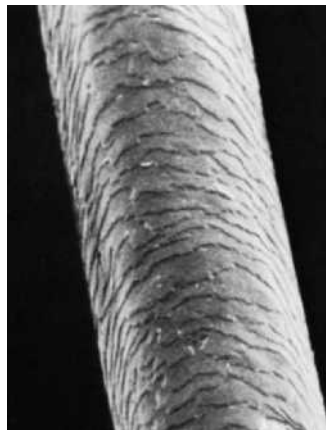
- Coarse hair has the largest diameter and tends to be stronger than fine hair. It also has a stronger structure, which may require more processing or stronger products during chemical services than medium or fine hair. Generally, it is more difficult for hair lighteners, haircolors, waving solutions, and relaxing creams to penetrate coarse hair (**Figure 10-21**).
- Medium hair texture is the most common and is the standard to which other hair is compared. Medium hair is the hair texture that is considered normal and does not usually pose any special problems or concerns during a haircut (**Figure 10-22**).
- Fine hair has the smallest diameter and is generally more fragile, easier to process, and more susceptible to damage from chemical services than coarse or medium hair. Some fine or very fine hair does not possess a medulla, which helps to account for the smaller diameter of the strand (**Figure 10-23**).

figure 10-21
Coarse hair.



The Gillette Research Institute.

figure 10-22
Medium hair.



The Gillette Research Institute.

figure 10-23
Fine hair.



The Gillette Research Institute.

- Wiry hair, whether coarse, medium, or fine, has a hard, glassy finish because the cuticle scales lie flat against the hair shaft. It usually takes longer to give this type of hair a chemical service.

Testing for Hair Texture

Hair texture can be determined by feeling a single strand of dry hair between the fingers. Hold the strand securely with one hand while rolling it between the thumb and forefinger of the other hand. With practice, you will be able to feel the difference between coarse, medium, and fine hair textures (Figure 10-24).

HAIR DENSITY

Hair density (HAYR DEN-sih-tee) refers to the number of individual hair strands per square inch of scalp area and is classified as thick, average, or thin or as high, medium, or low density.

- Hair density is different from hair texture in that individuals with the same hair texture can have different densities or different amounts of hair per square inch. For example, one person may have thick, fine hair while another has thin, fine hair.
- The average hair density is approximately 2,200 strands per square inch, with the average head containing about 100,000 individual strands.
- The number of hairs on the head varies with the color of the hair—blonds usually have the highest density and redheads the least density per square inch (Table 10-4).

HAIR POROSITY

Hair porosity (HAYR puh-RAHS-ut-ee) is the ability of the hair to absorb moisture. The degree of porosity is directly related to the condition of the cuticle layer of the hair. A compact cuticle layer is naturally more resistant to penetration, whereas porous hair has a raised cuticle layer that easily absorbs water. The porosity level of hair can be classified as low porosity, average porosity, and high porosity or as resistant, normal, and overly porous.

- Hair that has moderate or average porosity is considered normal (Figure 10-25). This hair type presents no special problems and chemical applications usually process as expected.
- Hair that is classified as having low porosity is considered resistant (Figure 10-26). This hair type absorbs the least amount of moisture and may require a more alkaline solution than other hair types.
- Hair with a high porosity level is considered overly porous and is usually the result of previous overprocessing (Figure 10-27). Hair in this condition is dry, damaged, fragile, and brittle. Although overly porous hair may absorb liquids quickly, the damaged and open cuticle scales may make it difficult for the hair to retain haircoloring pigments or to react favorably to chemical texture services. Reconditioning treatments, the use of low-pH products, and strands tests are recommended prior to performing chemical services on hair in this condition.

figure 10-24
Testing hair texture.



figure 10-25
Average porosity (normal hair).

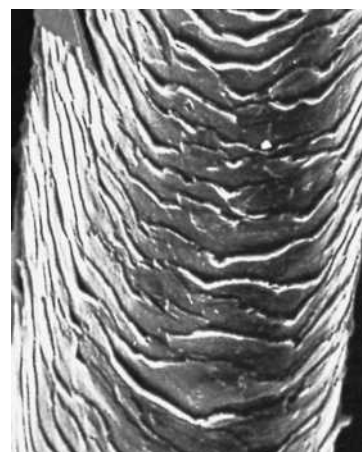


table 10-4
AVERAGE NUMBER OF
HAIRS ON THE HEAD
BY HAIR COLOR

Hair Color	Average Number of Hairs on Head
Blond	140,000
Brown	110,000
Black	108,000
Red	80,000

figure 10-26
Low porosity (resistant hair).



figure 10-27
High porosity (overly porous hair).

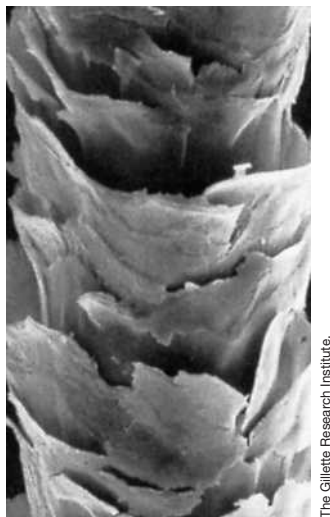


figure 10-28
Testing for hair porosity.



Testing for Hair Porosity

The porosity of the hair can be checked by holding multiple strands of dry hair between the fingers while sliding the thumb and forefinger of the other hand down toward the scalp (**Figure 10-28**). If the hair feels smooth and the cuticle is compact, it is considered resistant. If you can feel a slight roughness or the imbrications of the cuticle scales, the hair is porous. Should the hair break or feel very rough, the hair is probably over-porous.

HAIR ELASTICITY

Hair elasticity (HAYR ee-las-TIS-ut-ee) is the ability of the hair to stretch and return to its original length without breaking and is an indication of the strength of the side bonds in the hair. Hair with normal elasticity tends to hold the curl from sets and permanent waves without excessive relaxing of the curl. Hair with low or poor elasticity is brittle and breaks easily. Hair in this condition may have been overprocessed during previous chemical applications or may be the result of poor nutrition or internal disorders.

figure 10-29
Testing for hair elasticity.



Testing for Hair Elasticity

Gently tug a few strands of clean, wet hair as shown in **Figure 10-29**.

- Hair with normal elasticity will stretch and return to its original length without breaking.
- Curly or wavy hair may stretch up to 50 percent of its original length and return to that length without breaking.
- Hair that breaks easily or fails to return to its normal length has low elasticity.

REVIEW QUESTIONS

- 1 Why is the study of the properties and disorders of the hair and scalp important to barbers?
- 2 Name the five main structures of the hair root.
- 3 List the three layers of the hair shaft.
- 4 Describe the process of keratinization.
- 5 Describe polypeptide chains.
- 6 List and describe the three types of side bonds.
- 7 Name and describe the two types of melanin responsible for natural hair color.
- 8 Define and describe the two types of body hair and where they are found on the body.
- 9 List and describe the steps in the hair growth cycle.
- 10 Define and describe androgenic alopecia, alopecia areata, and alopecia totalis.
- 11 Name two FDA-approved treatments for hair loss.
- 12 Identify the hair and scalp disorders that must be referred to a physician for treatment.
- 13 Explain the similarities and differences associated with folliculitis barbae and pseudofolliculitis barbae.
- 14 Describe the factors to be observed and considered in a hair and scalp analysis.

CHAPTER GLOSSARY

alopecia (al-oh-PEE-shah)	p. 258	abnormal hair loss
alopecia areata (al-oh-PEE-shah ay-reh-AH-tuh)	p. 258	the sudden falling out of hair in patches or spots—autoimmune disorder
alopecia prematura (al-oh-PEE-shah pree-muh-TOO-ruh)	p. 258	hair loss that occurs before middle age
alopecia senilis (al-oh-PEE-shah seh-NIL-is)	p. 259	hair loss occurring in old age
alopecia syphilitica (al-oh-PEE-shah sif-il-IT-ih-kuh)	p. 258	hair loss as a result of syphilis
alopecia totalis (al-oh-PEE-shah toe-TAL-is)	p. 259	total loss of scalp hair
alopecia universalis (al-oh-PEE-shah yoo-nih-vur-SAA-lis)	p. 259	complete loss of body hair
amino acids (uh-MEE-noh AS-udz)	p. 250	the building blocks or units of structure in protein
anagen phase (AN-uh-jen FAYZ)	p. 256	growth phase in the hair cycle
androgenic alopecia (an-druh-JEN-ik al-oh-PEE-shah)	p. 258	hair loss that occurs as a result of genetics, age, and hormonal changes; male pattern baldness
arrector pili (ah-REK-tohr PY-leh)	p. 249	involuntary muscle fiber attached to the base of the hair follicle
canities (kah-NISH-ee-eez)	p. 260	technical term for gray hair
carbuncle (KAHR-bung-kul)	p. 265	the result of an acute, deep-seated bacterial infection in the subcutaneous tissue
catagen phase (KAT-uh-jen FAYZ)	p. 256	transition phase of the hair growth cycle that signals the end of the growth phase

cortex (KOR-teks)	p. 250	middle layer of the hair shaft
cowlick (KOW-lik)	p. 257	tuft of hair that stands straight up
cuticle (KYOU-tih-kul)	p. 249	outermost layer of the hair shaft
cysteine (SIS-tuh-een)	p. 252	an amino acid with a sulfur atom that joins together two peptide chains
dermal papilla (DUR-mul puh-PIL-uh)	p. 249	small, cone-shaped elevation located at the base of the hair follicle that fits into the hair bulb
disulfide bond (dy-SUL-fyed BAHND)	p. 252	also known as a <i>sulfur bond</i> ; a type of chemical cross bond found in the hair cortex
end bonds (END BAHNDZ)	p. 251	also known as <i>peptide bonds</i> ; chemical bonds that join amino acids end to end
eumelanin (yoo-MEL-uh-nin)	p. 253	melanin that gives brown and black color to hair
follicle (FAWL-ih-kul)	p. 248	tubelike depression in the skin that contains the hair root
folliculitis barbae (fah-lik-yuh-LY-tis BAR-bee)	p. 264	also known as <i>barber's itch</i> ; a bacterial infection of the hair follicles with inflamed pustules in the bearded areas of the face and neck; may have hairs growing through the pustule
fragilitas crinium (fruh-JIL-ih-tus KRY-nee-um)	p. 261	technical term for brittle hair
furuncle (FYOO-rung-kul)	p. 265	an acute bacterial infection of a hair follicle, producing constant pain; also known as a <i>boil</i>
hair bulb (HAYR BULB)	p. 249	club-shaped structure that forms the lower part of the hair root
hair density (HAYR DEN-sih-tee)	p. 267	the amount of hair per square inch of scalp
hair elasticity (HAYR ee-las-TIS-ut-ee)	p. 268	the ability of the hair to stretch and return to its original length
hair porosity (HAYR puh-RAHS-ut-ee)	p. 267	the ability of the hair to absorb moisture
hair root (HAYR ROOT)	p. 248	the part of the hair that is encased in the hair follicle
hair shaft (HAYR SHAFT)	p. 248	the part of the hair that extends beyond the skin
hair stream (HAYR STREEM)	p. 257	hair that flows in the same direction
hair texture (HAYR TEKS-chur)	p. 266	measures the diameter of a hair strand: coarse, medium, fine
helix (HEE-licks)	p. 251	the spiral shape of a coiled protein created when polypeptide chains intertwine
hydrogen bond (HY-druh-jun BAHND)	p. 252	weak, physical, cross-link side bond that is easily broken by water or heat
hypertrichosis (hy-pur-trih-KOH-sis)	p. 261	a condition of abnormal hair growth
keratin (KAIR-uht-in)	p. 248	the protein of which hair is made
keratinization (kair-uh-ti-ni-ZAY-shun)	p. 250	process by which protein cells mature within the follicle to form hair
lanugo (luh-NOO-goh)	p. 255	vellus hair
malassezia (mal-uh-SEEZ-ee-uh)	p. 261	fungus that causes dandruff
medulla (muh-DUL-uh)	p. 250	innermost or center layer of the hair shaft

melanin	p. 253	tiny grains of pigment in the cortex that give natural color to the hair
monilethrix (mah-NIL-ee-thriks)	p. 261	technical term for beaded hair
pediculosis capitis (puh-dik-yuh-LOH-sis KAP-ih-tus)	p. 263	the infection of the hair and scalp with head lice
peptide bonds (PEP-tyd BAHNDZ)	p. 251	end bonds; chemical bonds that join amino acids end to end
pheomelanin (fee-oh-MEL-uh-nin)	p. 253	melanin that gives red to blond colors to hair
pityriasis (pit-ih-RY-uh-sus)	p. 261	technical term for dandruff
pityriasis capitis simplex (pit-ih-RY-uh-sus KAP-ih-tus SIM-pleks)	p. 262	dry dandruff
pityriasis steatoides (pit-ih-RY-uh-sus stee-uh-TOY-deez)	p. 262	waxy or greasy dandruff
polypeptide chain (pahl-ee-PEP-tyd CHAYN)	p. 251	long chain of amino acids linked by peptide bonds
proteins (PROH-teenz)	p. 251	are made of chemical units called amino acids
pseudofolliculitis barbae (SUE-doe-fah-lik-yuh-LY-tis BAR-bee)	p. 264	a chronic inflammatory form of folliculitis known as <i>razor bumps</i> resembling folliculitis papules and pustules; generally accepted to be caused by ingrown hair
salt bond (SAWLT BAHNDZ)	p. 252	a physical side bond within the hair cortex
scutula (SKUT-yoo-lah)	p. 263	dry, sulfur-yellow, cuplike crusts on the scalp seen in tinea favosa
side bonds (SYD BAHNDZ)	p. 251	also known as <i>cross bonds</i> ; hydrogen, salt, and sulfur bonds in the hair cortex
sycosis vulgaris (sy-KOH-sis vul-GAYR-is)	p. 264	chronic bacterial infection of the bearded areas of the face
telogen phase (TEL-uh-jen FAYZ)	p. 256	resting phase of the hair growth cycle
terminal hair (TUR-mih-nul HAYR)	p. 255	long hair found on the scalp, beard, chest, back, and legs
tinea (TIN-ee-uh)	p. 262	technical name for ringworm
tinea barbae (TIN-ee-uh BAR-bee)	p. 263	also known as barber's itch, a superficial fungal infection that commonly affects the skin; it is primarily limited to the bearded areas of the face and neck or around the scalp
tinea capitis (TIN-ee-uh KAP-ih-tus)	p. 262	a fungal infection of the scalp characterized by red papules, or spots, at the opening of the hair follicles
tinea favosa (TIN-ee-uh fah-VOH-suh)	p. 263	ringworm characterized by dry, sulfur-yellow crusts on the scalp
tinea sycosis (TIN-ee-uh sy-KOH-sis)	p. 265	ringworm of the bearded areas on the face
trichology (trih-KAHL-uh-jee)	p. 248	the science dealing with the hair, its diseases, and its care
trichoptilosis (trih-kahp-tih-LOH-sus)	p. 261	the technical term for split ends; hair conditioning and oils will smooth split ends, but the only way to remove them is to cut them
trichorrhexis nodosa (trik-uh-REK-sis nuh-DOH-suh)	p. 261	technical term for knotted hair
vellus hair (VEL-us HAYR)	p. 255	soft, downy hair that appears on the body
wave pattern (WAYV PAT-URN)	p. 253	amount of movement in the hair strand; straight, wavy, curly, and coiled
whorl (WHORL)	p. 257	hair that grows in a circular pattern

PART 3 THE PRACTICE OF BARBERING

11 TREATMENT OF THE
HAIR AND SCALP

12 MEN'S FACIAL MASSAGE
AND TREATMENTS

13 SHAVING AND FACIAL-
HAIR DESIGN

14 MEN'S HAIRCUTTING
AND STYLING

15 MEN'S HAIR REPLACEMENT





11



TREATMENT OF THE
HAIR AND SCALP

CHAPTER OUTLINE

INTRODUCTION

WHY STUDY TREATMENT OF THE HAIR AND SCALP?

DISCUSS THE SHAMPOO SERVICE

KNOW HOW TO DRAPE

UNDERSTAND THE SHAMPOO SERVICE

LEARN ABOUT SCALP AND HAIR TREATMENTS

PROCEDURES 11-1 TO 11-7

LEARNING OBJECTIVES

After completing this chapter, you will be able to:

LO1

Discuss the benefits of a shampoo service.

LO2

Select products for different hair types and textures.

LO3

Describe proper draping procedures for various services.

LO4

Identify basic considerations for performing a shampoo service.

LO5

Describe two shampooing methods.

LO6

Discuss reasons why a client may find fault with a shampoo service.

LO7

Describe scalp massage manipulations and techniques.

LO8

Explain services that may be included in a hair or scalp treatment.

Introduction

The treatment of the hair and scalp includes shampooing, scalp massage, conditioning, and special treatments for hair and scalp conditions. When performed correctly, these services are relaxing and effective in helping to ensure the health of clients' hair and scalp. Providing such professional services can also increase client retention and referrals and helps to promote a positive reputation for the barber and barbershop.

why study

TREATMENT OF THE HAIR AND SCALP?

Barbers should study and have a thorough understanding of treatments for the hair and scalp because:

- Analyzing the client's hair and scalp helps to identify conditions that may prohibit services.
- Professionally delivered scalp massage during shampooing and treatments provides hygienic, circulatory, and relaxation benefits to the client.
- Special services, such as electrotherapy or light therapy treatments, require advanced knowledge and skills to perform correctly and safely.

Discuss the Shampoo Service

After reading this section, you will be able to:

- LO 1 Discuss the benefits of a shampoo service.
- LO 2 Select products for different hair types and textures.

figure 11-1
The shampoo service.



Shampooing the hair before cutting ensures that you are working with clean hair that is free from oils or products that can interfere with cutting tools and haircut results. Follow-up conditioning treatments after the shampooing help to keep hair in a healthy and manageable condition for the client and the barber (Figure 11-1). It is your responsibility to know how shampoos and conditioners affect the hair and scalp so that you can select or recommend the right products for your clients. As a professional barber, you need to take the time to read the specific information found on product labels or manufacturers' websites. Table 11-1 shows a general guideline for selecting shampoos and conditioners for different hair types and textures.

REVIEW SHAMPOOS AND CONDITIONERS

The purpose of a shampoo product and service is to cleanse the scalp and hair. This may seem obvious, but some barbers still encounter clients who use bar soap or other detergent products, which can leave the hair dry or coated with soap residues. As you may know from your study of chemistry,

table 11-1

MATCHING PRODUCTS TO HAIR TYPES

Hair Type	Hair Texture		
	Fine	Medium	Coarse
Straight	Volumizing shampoo Detangler, if necessary Protein treatments Fine-hair shampoo	pH-balanced shampoo Finishing rinse Protein treatments	Moisturizing shampoo Leave-in conditioner Moisturizing treatments
Wavy, curly, extremely curly	Light leave-in conditioner Protein treatments Spray-on thermal protectors	pH-balanced shampoo Leave-in conditioner Moisturizing treatment	Leave-in conditioner Protein and moisturizing treatments
Dry and damaged (perms, color-treated, relaxers, blowdrying, sun, hot irons)	Gentle cleansing shampoo Light leave-in conditioner Protein and moisturizing repair treatments Spray-on thermal protection	Shampoo for chemically treated hair Moisturizing conditioner Protein and moisturizing repair treatments	Deep-moisturizing shampoo for damaged hair Leave-in conditioner Deep-conditioning treatments and hair masks

shampoos (sham-POOz) are oil-in-water emulsions with a pH range of 4.5 to 7.5. Since these products do not contain the harsh alkalis found in soaps and detergents, they leave the hair in a more manageable condition.

Conditioners can refer to either hair conditioners or scalp conditioners.

- Generally, **hair conditioners** (HAYR kun-DIH-shun-urz) moisturize the hair and help to restore some of the oils and/or proteins. Hair conditioners typically range from 3.0 to 5.5 on the pH scale.
- **Scalp conditioners** (SKALP kun-DIH-shun-urz), usually in cream or ointment form, are available for overall scalp maintenance or to treat conditions requiring a medicinal product.

**REMINDER**

Display and use the shop's retail products at the workstation or shampoo sink to promote product sales and your endorsement of the products.

Know How to Drape

After reading this section, you will be able to:

LO3 Describe proper draping procedures for various services.

Before any barbering service can begin, the client needs to be appropriately draped for the service. **Draping** is the term used to describe the covering of the client's skin and clothing with a cape and a barrier at the neckband for their protection. Proper draping for the service being performed is required to maintain compliance with state barber laws and infection control standards in the barbershop.

Depending on the procedure to be performed, you will drape a client for either a wet service or a dry service. Wet services, such as shampooing

figure 11-2
Draped shampoo cape.



figure 11-3
Draped haircutting cape.



REMINDER

If a shampoo service precedes the haircut, replace the vinyl cape and towel with a nylon cape and neck strip.

DID YOU KNOW?

Paper towels designed for barbering services are available and preferred by some barbers. They typically measure 12 × 24 inches and can be used

- in place of a neck strip;
- as a wiping towel for lather during razor work;
- as a towel wrap over the hand to remove hair clippings or to apply products.

and chemical applications, require protection from liquids. Dry services, such as a haircut or beard trim, do not require this type of protection, so it is important to use the appropriate drape for the service.

The two main types of drapes used are shampoo capes (or wet capes) and haircutting capes (also known as *chair cloths*).

- *Shampoo capes* are waterproof drapes made of vinyl that are used to protect the client's skin and clothing from water, liquids, and chemical processes (see **Figure 11-2**).
- *Haircutting capes* are made of nylon or other synthetic materials. These draping fabrics are usually more comfortable for the client because they do not hold in as much body heat as vinyl capes. From the barber's standpoint, these fabrics are also more effective for shedding wet and dry hair. Wet hair has a tendency to stick to vinyl capes, making it more difficult to shake loose hairs off the drape (**Figure 11-3**).

LEARN DRAPING METHODS

The method of draping to be used depends on the service to be performed. There are several draping methods used in the barbershop and all require the use of a barrier (towel or neck strip) between the client's neck and the neckband of the cape. The barriers used are listed below; review Procedures 11-1 and 11-2 for guidelines on draping for wet, chemical, and haircutting services.

- **Shampoo service.** A cloth towel is positioned under the cape and folded over the neckband once the cape is secured. Another option for the shampoo service is to place a second towel over the cape neckline, secured with a clip, to use for blotting the hair after the shampoo procedure.
- **Chemical service.** A chemical service always requires a cloth towel under the cape and folded over the neckband once the cape is secured, followed by a cloth towel over the cape neckline and secured with a clip to provide protection from solution or chemical drips.
- **Haircut service.** For a haircut service, draping includes the use of a neck strip under and then folded over the neckband of the cape.
- **Mustache/beard trim service.** This service requires either a neck strip or a cloth towel placed under and then folded over the neckband of the cape. Some barbers prefer to use a terry cloth towel for beard trims because the cloth fibers are more effective in trapping hair clippings than a neck strip.
- **Shave service.** The shave service requires a special draping method to ensure that a barrier between the client's skin and the drape remain in place during the procedure. This method is described in detail in Chapter 13.

P 11-1 Draping for Wet and Chemical Services *pages 286–287*

P 11-2 Draping for Haircutting Services *pages 288–289*

Although several draping methods are presented in this text, your instructor's methods would be correct as well, as long as there is a barrier between the client's neck and the neckband of the cape.

Consider Important Guidelines for Draping

Important guidelines that apply to all draping methods are as follows:

- Prepare materials and supplies for the service.
- Wash your hands.
- Ask the client to remove jewelry if any and store it in a safe place.
- Turn the client's collar to the inside if applicable (Figure 11-4).
- Proceed with the appropriate draping method.

Understand the Shampoo Service

After reading this section, you will be able to:

- LO 4 Identify basic considerations for performing a shampoo service.
- LO 5 Describe two shampooing methods.
- LO 6 Discuss reasons why a client may find fault with a shampoo service.

The basic considerations for performing a shampoo service include proper draping and positioning of the client, scalp manipulations to facilitate the shampoo procedure, and proper body positioning of the barber. Most barber shops are equipped with shampoo bowls either within a working booth area or placed in a separate section of the shop. Typically, the barber stands beside the client and shampoo bowl while performing the shampoo. Some shops, however, are equipped with the European-style shampoo bowl, which is a freestanding unit that allows the barber to stand in back of the client's head.

DESCRIBE TWO METHODS OF SHAMPOOING AND RINSING

The two methods used for shampooing and rinsing are the reclined and inclined methods.

- The *reclined method* of shampooing is the most commonly used method. The hydraulic or shampoo chair is reclined and the client's head positioned in the neck rest of the shampoo bowl (Figure 11-5). This method is favored because it is more comfortable for the client and permits greater speed and efficiency by the barber.
- The *inclined method* can be used when a standard shampoo bowl is not available or when the client cannot use the reclined method. This method requires the client to bend his head forward over the shampoo bowl or sink (Figure 11-6). The client may also sit on a stool or chair positioned close to the sink.

SHAMPOOING CLIENTS WITH SPECIAL NEEDS

If a client is disabled or wheelchair bound, ask the client how he would like to be shampooed. Some wheelchairs can be positioned comfortably at the shampoo bowl while others may require the client to use the inclined method. Always make the client's comfort and safety a priority.

figure 11-4

Turn client's collar to the inside.



DID YOU KNOW?

The purpose of the towel or neck strip in draping procedures is to prevent the cape from having direct contact with the client's skin and to maintain compliance with state barber laws and infection control standards. This barrier is a requirement of every state's barber rules and regulations.

figure 11-5

Reclined method.



figure 11-6

Inclined method.



The Barber's Physical Presentation

To prevent muscle aches, back strain, and fatigue, it is important that you maintain good posture at the shampoo bowl. Review the following suggestions to maintain a good posture while shampooing.

- Stand as close as possible to the back of the client's head.
- Flex the knees slightly and position your body directly over your feet to maintain good balance.
- Try to keep your chin parallel to the floor to avoid neck strain. Your head should be raised, with the chest up, abdomen flat, and shoulders relaxed.
- Do not bend or twist sideways from the waist or lean too far forward.

Perform a Superior Shampoo Service

An excellent shampoo service requires you to give individual attention to each client's needs. The effectiveness of the shampoo service will depend on the manner in which you apply and rinse the shampoo, the quality of the scalp massage, the temperature of the water used, and the use of the shampoo best suited to the condition of the client's scalp and hair.

Some of the more common reasons that a client may find fault with the shampoo service include

- improper shampoo selection;
- insufficient scalp massage;
- extreme water temperatures: either too hot or too cold;
- shampoo or water that runs onto the client's face, ears, or eyes;
- wetting or soiling the client's clothing;
- scraping or scratching the client's scalp with fingernails;
- improper hair blotting;
- insufficient cleansing and rinsing.

An excellent shampoo service therefore requires preparation and set-up, product selection, water temperature testing, shampoo application, and shampoo massage manipulations.

Preparation and Setup

Following consultation with the client, preparation is the first step in performing a shampoo service. Assemble all necessary supplies or have them stored in an area convenient to the shampoo bowl. Supplies include cleansing and conditioning products, waterproof drapes, and terry cloth towels.

Analysis and Product Selection

It is essential that you be knowledgeable about the products you use on your clients. Always read product labels and follow the manufacturer's directions. To determine which product to use, you must first perform a scalp and hair analysis (**Figure 11-7**). Six characteristics of the scalp and hair that should be considered before choosing products are:

- **Condition of the scalp.** Dry, oily, normal, and/or the presence of abrasions or disorders such as tinea or pediculosis capitis (if abrasions or disorders are present, do not proceed with the service)

figure 11-7
Analyze the hair and scalp.

